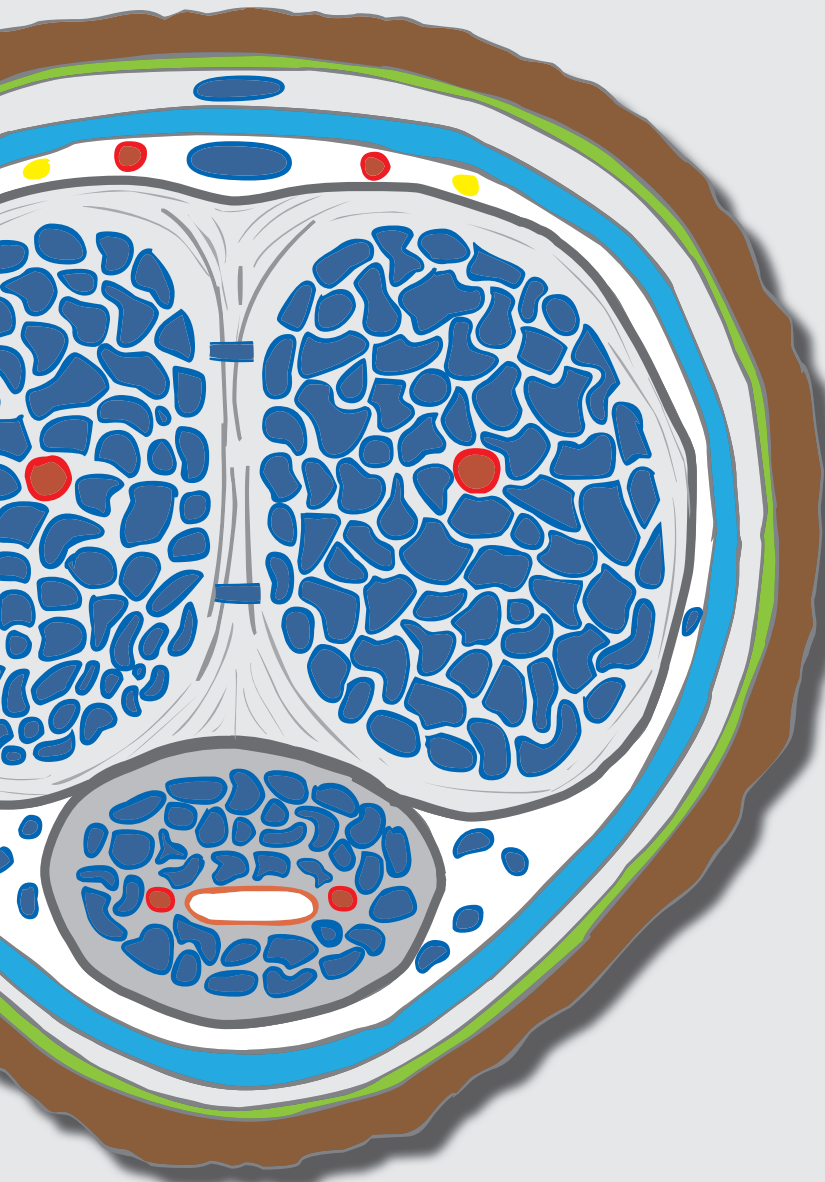


8

Genital system



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The sex of the foetus is **genetically determined at the moment of conception**. However, it is not until the **7th prenatal week** that it is **morphologically discernable**. The **process of differentiation** takes place until the 12th prenatal week, when the two sexes become fully differentiated. The organs of both the male and female genital systems can be divided into the gonads, excretory ducts and external genital organs.

Development of male sex organs

Development of female sex organs

1 Mesonephric duct of Wolff	2 Paramesonephric duct of Müller
Epididymis	Uterine tube
Ductus deferens	Uterus
Seminal glands	Cranial portion of vagina
3 Gonadal ridge	
Testis	Ovary
4 Primordia of prostatic buds but also urogenital sinus	
Prostate	Paraurethral glands
5 Urogenital sinus	
Prostatic and membranous urethra	Female urethra and vaginal vestibule
6 Bulbo-urethral gland primordium and greater vestibular gland primordium	
Bulbo-urethral gland	Greater vestibular gland
7 Phallic/genital tubercle	
Body of penis, glans penis and prepuce	Clitoris and prepuce of clitoris
8 Secondary urethral folds	
Root and raphe of penis	
Spongy urethra	Labia minora and bulb of vestibule
9 Labioscrotal/genital swellings	
Scrotum	Labia majora
10 Gubernaculum	
Gubernaculum testis → Scrotal ligament	Round ligament of uterus Proper ligament of ovary

The functional tissue of the ovary is its cortex (*cortex ovarii*). The functional tissue of the testis is its medulla with the seminiferous tubuli (*tubuli seminiferi*).

Remnants of the mesonephric tubules (*tubuli mesonephrici*) **in the female:** the ductulus transversus epoophori and paroophoron Kobelti

in the male: the superior and inferior aberrant ductule of Haller (*ductulus aberrans superior et inferior Halleri*) in the epididymis and the paradidymis of Giralde (*paradidymis Giraldesi*).

Remnants of the mesonephric duct in the female: the longitudinal duct of Gartner and Malpighi (*ductus longitudinalis epoophori Gartneri-Malpighi*) and the vestige of the ductus deferens (*ductus deferens vestigialis*)

in male: the appendix of the epididymis (*appendix epididymidis*).

Remnants of the paramesonephric duct

in the female: vesicular appendices of Morgagni (*appendices vesiculosae Morgagni*)

in the male: the appendix of the testis of Morgagni (*appendix testis Morgagni*)

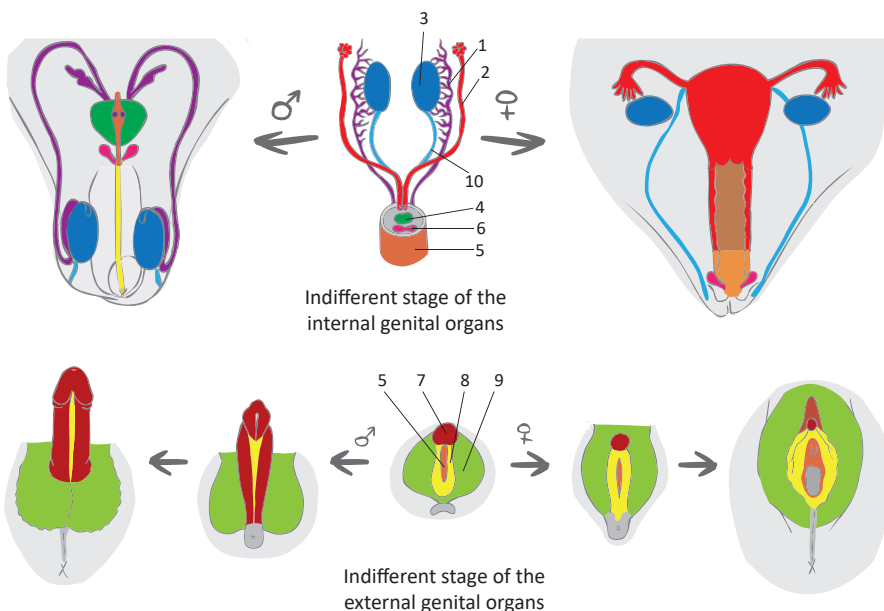
The paramesonephric duct disappears under the influence of anti-paramesonephric hormone (AMH/MIS, *factor antiparamesonephricus*).

Clinical notes

During the indifferent stage of development of the genital system, hormonal imbalance and other insults may lead to disorders of sex development, where phenotype does not correspond to sex genotype.

Embryological remnants of the genital system may give rise to cysts.

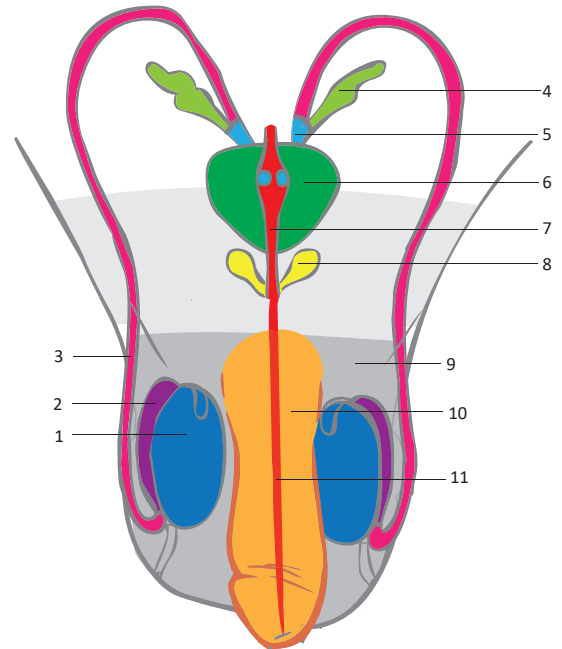
The appendix testis (hydatid of Morgagni) is a vestigial remnant of the paramesonephric duct, present at the upper pole of the testis. It can undergo a torsion, causing acute one-sided testicular pain.



The male genital organs are divided into the external and internal male genitalia according to the developmental origin from the urogenital sinus or urogenital ridge, respectively. **The testes, epididymides, ductus deferentes, seminal glands and prostate** belong to the internal male genital organs. **The scrotum and penis** constitute the external male genital organs. **The male urethra** is both an internal and external genital organ. Its cranial part is internal and its caudal part within the penis is external. As the testes are located **outside the peritoneal cavity**, they are kept at a slightly lower temperature of 32–34 °C, which is optimal for spermatogenesis.

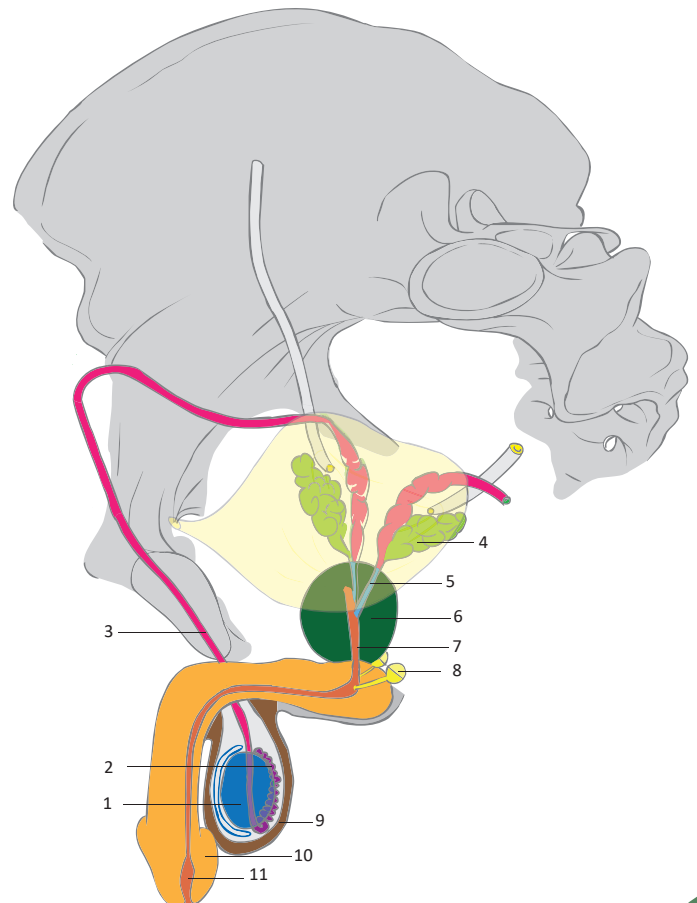
Internal genital organs

- **1 Testis**
 - a paired gland located inside the scrotum
 - produces spermatozoa and testosterone
- **2 Epididymis**
 - a paired organ serving as a place for maturation and storage of spermatozoa
- **3 Ductus deferens**
 - a paired duct that connects the epididymis to the urethra
- **4 Seminal gland (*glandula vesiculosa*)**
 - a paired organ lying on the posterior surface of the urinary bladder
 - produces most of the volume of the ejaculate
- **5 Ejaculatory duct (*ductus ejaculatorius*)**
 - a paired common duct formed by the ductus deferens and excretory duct of the seminal gland, terminates in the urethra
- **6 Prostate (*prostata*)**
 - an unpaired organ located under the urinary bladder
 - produces only a minor part of the ejaculate
 - the ejaculatory ducts terminate in the prostatic urethra
- **7 Male urethra (*urethra masculina*)**
 - the common terminal part of both the male genital and urinary systems
 - the internal part of the male urethra passes through the prostate and urogenital diaphragm
- **8 Bulbo-urethral glands (*glandulae bulbourethrales*)**
 - a pair of glands that empty into the urethra below the pelvic floor



External genital organs

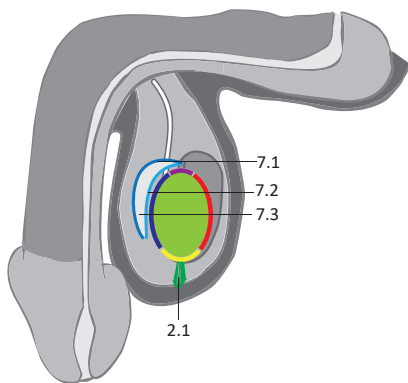
- **9 Scrotum**
 - a space outside the abdominopelvic cavity where the testis, epididymis and central part of the ductus deferens are located
- **10 Penis**
 - an unpaired organ that contains the terminal segments of both the urinary and genital systems
 - functions as an organ of micturition and copulation
- **11 Male urethra (*urethra masculina*)**
 - the common terminal part of both the male genital and urinary systems
 - the external part of the male urethra passes through the corpus spongiosum of the penis



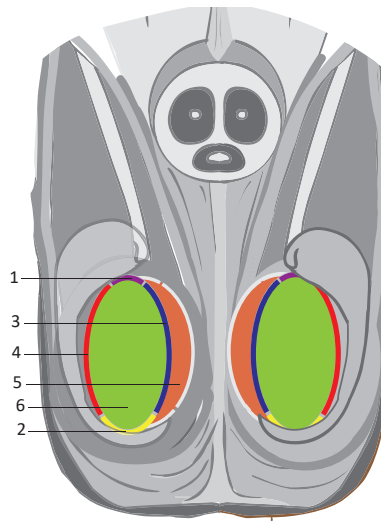
The testes are a pair of genital glands located outside the abdominopelvic cavity in the scrotum. During prenatal development, they originate in the lumbar region and descend through the retroperitoneal space and inguinal canal to reach the scrotum. The layers of the abdominal wall form the layers of scrotum as the testes passes through them. The testicular vessels descend with the testes, hence the origin of these arteries in lumbar region of the abdomen. Lymphatic drainage is similarly directed to the lumbar nodes. Testosterone and spermatozoa are produced in the testes.

External structure

- 1 Upper pole (*extremitas superior*)
- 2 Lower pole (*extremitas inferior*) – the attachment of the scrotal ligament to the testis
 - 2.1 Scrotal ligament (*ligamentum scrotale*) – a ligament attaching the lower pole of the testis and epididymis to the scrotum
- 3 Anterior border (*margo anterior*) – the anterior margin facing the tunica vaginalis
- 4 Posterior border (*margo posterior*) – the posterior margin facing the epididymis
- 5 Medial surface (*facies medialis*) – the medial surface facing the septum of the scrotum
- 6 Lateral surface (*facies lateralis*)
- 7 Tunica vaginalis (*tunica vaginalis testis*) – originally a part of peritoneum (*processus vaginalis peritonei*) that became one of the layers of the scrotum during the descent of the testis – is composed of two layers with a slender slit between them
 - 7.1 Parietal layer (*lamina parietalis / periorchium*)
 - 7.2 Visceral layer (*lamina visceralis / epiorchium*)
 - 7.3 Cavum serosum scroti – a slender cavity between the parietal and visceral layers of the tunica vaginalis



Sagittal section of the penis and scrotum



Frontal section of the penis and scrotum

Internal structure

Capsule

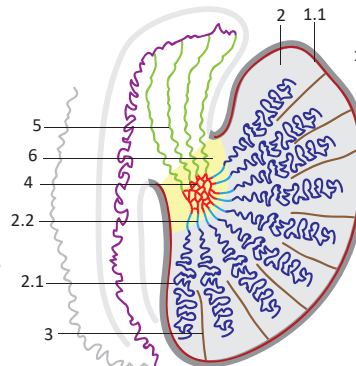
- 1 Tunica albuginea – a dense membrane of connective tissue covering the testis
 - 1.1 Vascular layer (*tunica vasculosa*) – is located under the tunica albuginea

Lobules

- 2 Lobules of testis (*lobuli testis*) – each contains a single tubule
 - 2.1 Convoluted seminiferous tubules (*tubuli seminiferi contorti*) – are lined with complex seminiferous epithelium that contains spermatogenic cells
 - 2.2 Straight tubules (*tubuli seminiferi recti*) – are continuous with the rete testis
- 3 Septa of testis (*septula testis*) – separates the lobules – dense plates of connective tissue derived from the tunica albuginea

Ducts

- 4 Rete testis – a network of tubules in the mediastinum testis
- 5 Efferent ductules (*ductuli efferentes testis*) – connect the retes testis to the epididymis – are located in the head of the epididymis
- 6 Mediastinum of the testis (*mediastinum testis*) – an area close to the hilum of the testis filled with connective tissue where the rete testis and efferent ductules coalesce



Orchis is the Greek term for testis.

The gubernaculum testis is a thick band of connective tissue that originates from the caudal part of the urogenital fold (*plica urogenitalis*). It is attached to the mesenchyme of the genital swellings on both sides of the urogenital membrane. Due to its firm attachment to the scrotum, it directs the testis during its descent.

The tunica albuginea is thickened on the posterior surface of the testis where it pierces the parenchyma of the testis and forms the mediastinum testis.

The hilum of the testis is a slit-like area on the posterior surface of the testis where the testicular vessels enter and the efferent ductules leave the testis.

There are 200–300 lobules in the testis.

Layers of the scrotum – page 246.

Clinical notes

Due to the origin of the testis in the retroperitoneal space, it is necessary to check the retroperitoneal lymph nodes for potential metastases as a part of surgical treatment of testicular tumors.

Palpation of the testes in the scrotum is an important part of the physical examination in paediatrics. The absence of the testes in situ, enlargement, pain or other physical abnormalities may be a sign of a potentially serious illness.

Testicular torsion describes an acute situation when the testis twists around the vessels passing in the spermatic cord. The symptoms are swelling of the scrotum, intense pain projecting to the inguinal region and nausea. It usually occurs in adolescent males and requires urgent surgical treatment to prevent complete necrosis of the testis. After 12 hours, the testis is usually irreversibly damaged and it is necessary to remove it.

Orchidectomy is a surgical removal of the testis.

Orchidopexy describes a surgical treatment of cryptorchism. It is a procedure to move an undescended testis to its proper place in the scrotum. Orchidopexy should be performed within the first six years of life, otherwise the risk of infertility and testicular tumors increases.

Descent of the testis

Descent of the testis from their place of their origin in the lumbar region to the scrotum takes place during intrauterine development:

- 3rd month** – unequal growth of the trunk and the pulling force of the gubernaculum begins to move the testis caudally from their position at L1–L2
 - the testicular artery (from the aorta), testicular vein (to the inferior vena cava) and lymphatic vessels follow the testis during its descent
 - the testis is located in the iliac fossa by the end of the 3rd month
- 4th month** – the testis is located in the deep inguinal ring
- 6th month** – the testis passes through the inguinal canal and pulls the peritoneum (which becomes the vaginal process) fasciae of the abdominal muscles and muscle bundles of the transversus abdominis and internal oblique along with it
 - these structures become the layers of the scrotum and the cremaster muscle
 - vessels, nerves and the ductus deferens form the contents of the spermatic cord
- 8th month** – the testis should be fully descended
 - the arrangement of the scrotal layers are fully formed
 - the vaginal process of the peritoneum separates completely from the abdominal peritoneum and closes to form the tunica vaginalis

Descended testes (testes in situ) is an important sign of maturity of the newborn. Spermatogenesis is only possible in testes positioned inside the scrotum. Descent of the testes should be complete by the end of the **3rd postnatal month**.

Innervation

Sympathetic system: the testicular plexus (located around the testicular artery)

Parasympathetic system: the vagus nerve

Visceral sensory system: thoracic nerves (that run to spinal cord segment T10)

Vascular supply

Arterial supply:

- 1 **Testicular artery** (from the abdominal aorta), partially: artery to ductus deferens (from the internal iliac artery) and cremasteric artery (from the inferior epigastric artery)

Venous drainage:

- 2 **Pampiniform plexus and then testicular veins**
 - 2.1 **Right testicular vein** – to the inferior vena cava
 - 2.2 **Left testicular vein** – to the left renal vein

Lymphatic drainage:

- 3 **Lumbar nodes**

Function

Spermatozoa production occurs in the seminiferous epithelium.

Sertoli cells support, isolate and supply spermatozoa precursor cells (spermatogonia, spermatocytes and spermatids) with nutrients.

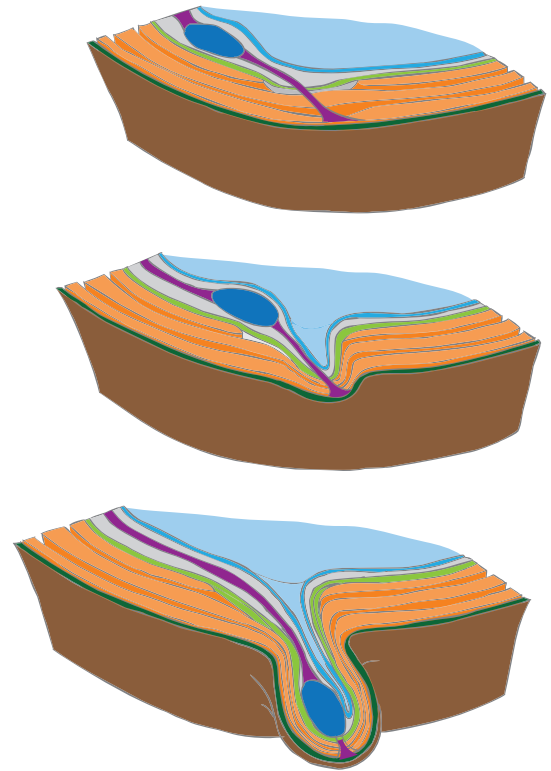
They also maintain high levels of testosterone in the seminiferous epithelium, which is necessary for the proper development of spermatozoa.

Testosterone production takes place in Leydig cells, located in the interstitium of the seminiferous tubules. Testosterone is the main male sex hormone (androgen). It is responsible for male primary and secondary sex characteristics.

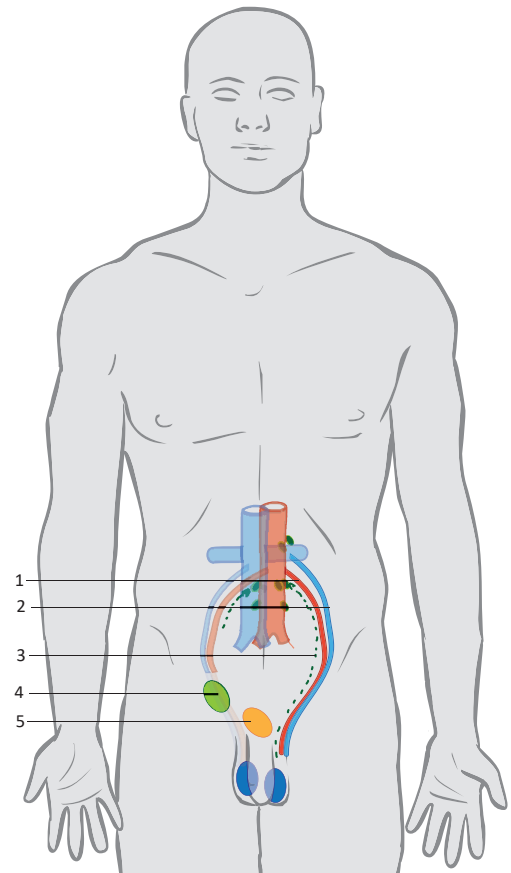
Clinical notes

Cryptorchism (undescended testis, retention of the testis)

- failure of descent of the testis and its subsequent absence in the scrotum
- 4 **Dystopic testis** – retention of the testis anywhere on its course from the abdomen at level of L1–2 to the scrotum
 - is most commonly located within the inguinal canal
- 5 **Ectopic testis** – a testis located in a position outside of the course of its physiological descent to the scrotum (e.g. lesser pelvis, perineum, above the pubic symphysis)



Descent of the testis in the 3rd, 4th and 8th month of development



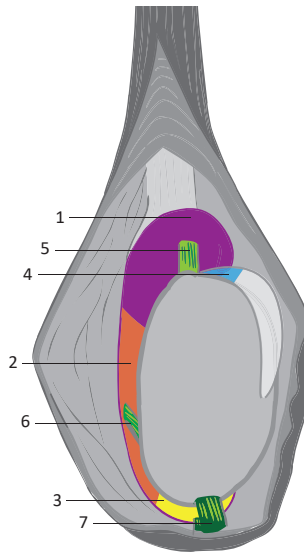
The epididymis is a **highly convoluted duct**. It is attached to the posterior margin of the testis and inferiorly to the scrotum. The tail of the epididymis **continues as the ductus deferens**. Spermatozoa spend 10–12 of their total 72 days of maturation within the epididymis. The epididymis serves as a **reservoir of spermatozoa**, where they mature and gain the capacity to move and fertilise the ovum. It also provides nutrition for the spermatozoa.

External structure

- **1 Head of epididymis** (*caput epididymidis*)
 - the cranial part of the head of the epididymis is formed by the efferent ductules
- **2 Body of epididymis** (*corpus epididymidis*)
 - is formed by the convoluted duct of the epididymis
- **3 Tail of epididymis** (*cauda epididymidis*)
 - the caudal part of the epididymis
 - connects to the ductus deferens
- **4 Sinus of epididymis** (*sinus epididymidis*)
 - a peritoneal fold
 - the visceral layer of the tunica vaginalis (*tunica vaginalis testis*) between the testis and epididymis

Supporting ligaments

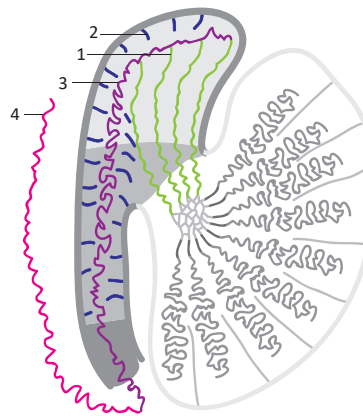
- **5 Superior ligament of epididymis** (*ligamentum epididymidis superius*)
 - connects the upper pole of the testis to the epididymis
- **6 Inferior ligament of epididymis** (*ligamentum epididymidis inferius*)
 - connects the lower pole of the testis to the epididymis
- **7 Scrotal ligament** (*ligamentum scrotale*)
 - connects the epididymis and testis to the caudal aspect of the scrotum
 - a remnant of the gubernaculum testis



Sagittal view of a dissected scrotum

Ducts and lobules

- **1 Efferent ductules** (*ductuli eferentes*)
 - 8–12 ductules that pass from the testis to the duct of the epididymis
- **2 Lobules of epididymis** (*lobuli epididymidis*)
 - each lobule is formed by a single highly convoluted efferent ductule
- **3 Duct of epididymis** (*ductus epididymidis*)
 - a highly convoluted duct of the epididymis, where spermatozoa definitively mature
- **4 Ductus deferens** – a duct connecting the duct of the epididymis with the prostate and urethra



Vascular supply

Arterial supply: the testicular artery (from the abdominal aorta)

Venous drainage: the pampiniform plexus (to the testicular veins, which drain to the inferior vena cava on the right side and to the left renal vein on the left side)

Lymphatic drainage: the lumbar nodes

Innervation

Sympathetic system: the testicular plexus (along the testicular artery)

Parasympathetic system: the vagus nerve

Visceral sensory: spinal nerves from the segments T11–T12

If uncoiled, the ductules of the epididymis would reach a length of six meters.

The **vas efferens** is an obsolete synonym for the ductus deferens.

The **pampiniform plexus** and the testicular veins work as a counter-current blood cooling system.

The **epithelium of the efferent ductules** is simple cuboidal with variable cell height. It contains kinocilia.

The **epithelium of the duct of the epididymis** is pseudostratified columnar with stereocilia.

Clinical notes

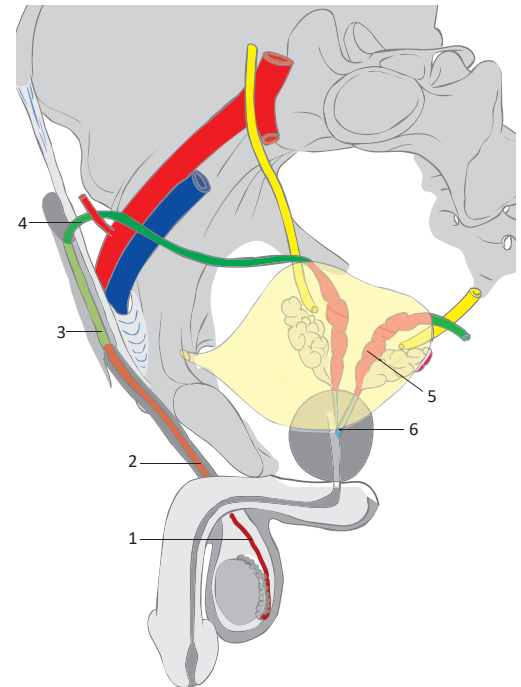
Varicocele is a term describing dilation of the pampiniform plexus inside the spermatic cord. In young males, the cause remains unknown. In older men, varicocele may arise as a result of impeded blood flow caused by various renal diseases and tumors. It is one of the most common causes of male infertility, as the impeded blood flow results in an increase in temperature inside the testis. Varicocele occurs more commonly on the left side, since the left testicular vein drains into the left renal vein perpendicularly.

The **appendix of the epididymis** (pedunculated hydatid) is a small stalked sac on the head of the epididymis. It is derived from the mesonephric duct.

The ductus deferens is a tubular organ **that transports spermatozoa from the epididymis to the prostatic urethra** by peristaltic contractions. It passes **from the scrotum through the spermatic cord and inguinal canal** into the lesser pelvis, where it **crosses the ureter** and **enters the prostate**.

Parts

- **1 Scrotal part** (*pars scrotalis*) – is located in the scrotum,
 - is continuous with the duct of the epididymis
- **2 Funicular part** (*pars funicularis*)
 - is located in the spermatic cord (*funiculus spermaticus*)
- **3 Inguinal part** (*pars inguinalis*)
 - is located in the inguinal canal (*canalis inguinalis*)
 - the ductus deferens enters the inguinal canal through the superficial inguinal ring, courses through the canal ventral to the inferior epigastric vessels and enters the pelvis through the deep inguinal ring
- **4 Pelvic part** (*pars pelvica*)
 - courses through the lesser pelvis, close to the urinary bladder and fuses with the excretory duct of the seminal vesicle
 - **External iliac vessels** – pass posteriorly to the ductus deferens
 - **Ureter** – runs beneath the ductus deferens in close proximity to the fundus of the urinary bladder
- **5 Ampulla of ductus deferens** (*ampulla ductus deferentis*)
 - the dilated, most distal segment of the ductus deferens
- **6 Ejaculatory duct** (*ductus ejaculatorius*)
 - a common duct formed by the union of the ductus deferens with the excretory duct of the seminal vesicle
 - enters the prostate and joins the prostatic urethra at the seminal colliculus



Left superior-oblique view of the male pelvis

2.4

Spermatic cord – Funiculus spermaticus

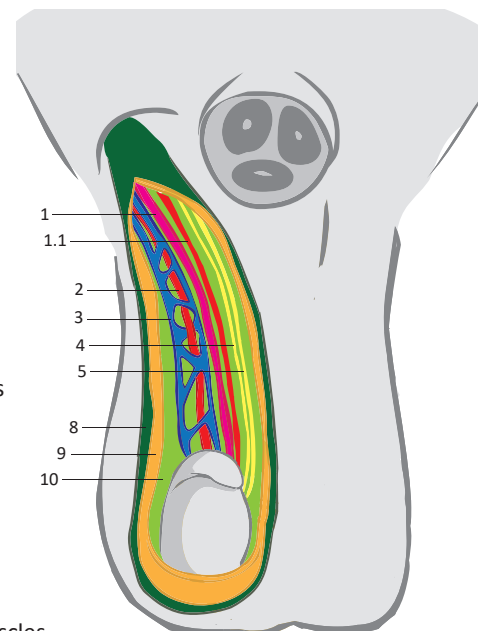
The spermatic cord is a **tough rope-like structure that contains the ductus deferens and several vessels and nerves**. It courses from **the scrotum to the inguinal canal**. The venous pampiniform plexus lowers the temperature inside the spermatic cord, which is necessary for the proper maturation of spermatozoa. The layers of the abdominal wall continue over the spermatic cord and the scrotum, forming their coverings.

Contents

- **1 Ductus deferens** – runs from the epididymis to the prostate
 - **1.1 Artery to ductus deferens** (*arteria ductus deferentis*)
 - a branch of the internal iliac artery that supplies the ductus deferens
- **2 Testicular artery** – a branch of the abdominal aorta supplying the testis and the epididymis
- **3 Pampiniform plexus** (*plexus pampiniformis*) – a venous plexus conveying blood from the testis and epididymis to the inferior vena cava on the right and to the left renal vein on the left
- **4 Testicular plexus** (*plexus testicularis*) – the nervous plexus of the testis
- **5 Deferential plexus** (*plexus deferentialis*)
 - a visceral sensory nervous plexus of the ductus deferens
- **6 Lymph vessels** – lymph from the testis and epididymis flows to the lumbar lymph nodes
- **7 Vestige of processus vaginalis** (*vestigium processus vaginalis testis*)
 - a remnant of the peritoneal fold

Coverings

- **8 External spermatic fascia** (*fascia spermatica externa*)
 - a continuation of the superficial abdominal fascia
- **9 Cremaster** – a skeletal muscle that pulls the scrotum towards the abdominal wall
 - consists of muscle bundles from the internal oblique and transversus abdominis muscles
 - the cremasteric fascia (*fascia cremasterica*) covers the cremaster and its muscle bundles
- **10 Internal spermatic fascia** (*fascia spermatica interna*) – a continuation of the transversalis fascia of the ventrolateral abdominal wall



Frontal section of the penis and scrotum

The seminal glands are **located behind the urinary bladder above the prostate**. Each seminal gland is a simple tubular gland, **composed of a single duct that has multiple convolutions**. The seminal glands produce **50–80 % of the ejaculate**. The excretory duct of the seminal gland joins the ductus deferens to form **the ejaculatory duct** that empties into the urethra.

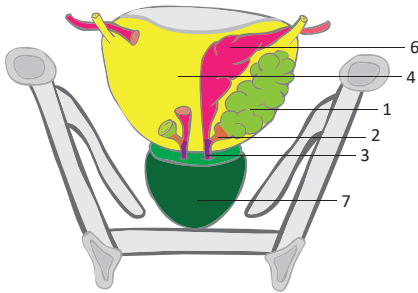
Regional anatomy and ducts

Seminal gland and its ducts

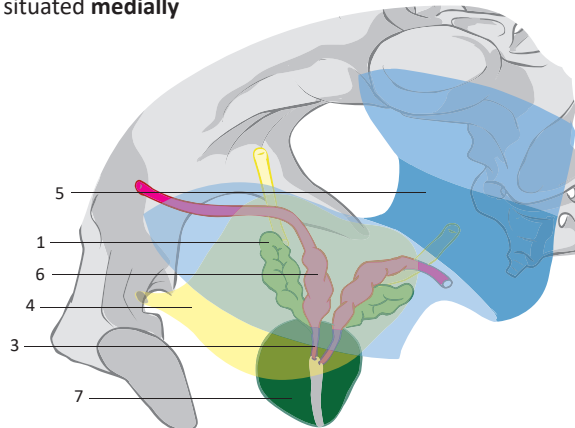
- 1 **Seminal gland** (*glandula vesiculosa*)
- 2 **Excretory duct** (*ductus excretorius*) – a duct of the seminal gland joining the ductus deferens
- 3 **Ejaculatory duct** (*ductus ejaculatorius*) – a common duct formed by the union of the excretory duct and ductus deferens

Regional anatomy

- 4 The posterior wall of the urinary bladder lies **anteriorly**
- 5 The rectovesical pouch of the peritoneal cavity is located **posteriorly**
- 6 The ampulla of the ductus deferens is situated **medially**
- 7 The prostate is found **caudally**



Posterior view of the seminal glands



Left upper oblique view of the male pelvis

Vascular supply

Arterial supply: artery to ductus deferens, inferior vesical artery, middle rectal artery

Venous drainage: prostatic venous plexus, vesical venous plexus

Lymphatic drainage: internal and external iliac lymph nodes

Innervation

Sympathetic system: prostatic plexus derived from the inferior hypogastric plexus

Parasympathetic system: prostatic plexus derived from the inferior hypogastric plexus

Viscerosensory fibres: are located in the prostatic plexus

Glandulae seminales, vesiculae seminales, seminal vesicles and vesicular glands are alternative terms for the seminal glands.

Verumontanum is an obsolete term for the seminal colliculus.

Seminal glands produce an alkaline substance that contains fructose (a nutrient for spermatozoa), globulin proteins and prostaglandins (hormone-like compounds that stimulate the smooth muscle of the female genital organs).

The seminal glands develop from the ductus deferens. They are 4–5 cm long, 17–20 mm wide and 6–9 mm thick. When uncoiled, the length of the simple tubular gland is 10–15 cm.

The seminal glands are lined with pseudostratified columnar epithelium.

The prostatic glands are lined with cuboidal to pseudostratified columnar epithelium (depending on the secretion activity of the gland).

Proportion of gland tissue in the different histological zones of the prostate: Peripheral zone 75 %
Central and transitional zone 20 %
Aglandular zone (isthmus) and periurethral zone 5 %.

Clinical notes

Per rectum palpation of the prostate is a standard part of the clinical examination, performed by palpating the prostate through the anterior wall of the rectum.

The zones of the prostate vary in shape, in proportion of fibromuscular stroma and in their tendency to develop disease. The different zones of the prostate can be visualised by transrectal ultrasound and distinguish in the magnetic resonance imaging.

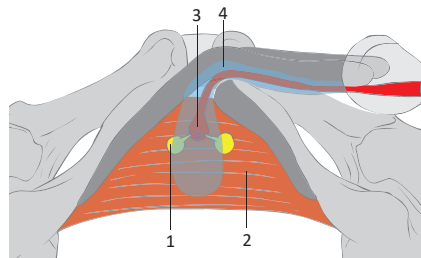
Benign prostate hyperplasia is characterised by an increase in number of the submucosal prostate glands. This leads to the enlargement of transitional zone, which can lead to obstruction of urinary flow. The enlarged transitional zone compresses the fibromuscular stroma of the peripheral zone, eventually transforming it into the “surgical” capsule of the prostate. This allows surgical enucleation of transitional zone, sparing the peripheral part.

Carcinoma of the prostate is a malignant neoplasia arising in the glandular epithelium of the peripheral zone or, less commonly, in the central or transitional zones.

The bulbourethral glands are a pair of **accessory genital glands that empty into the ampulla urethrae**. These pea-sized glands are located on the inferior surface of the deep transverse perineal muscle. They produce a viscous liquid that lubricates the internal surface of the urethra during ejaculation.

The gland and its regional anatomy

- 1 **Bulbo-urethral gland** (*glandula bulbourethralis*)
- 2 **Superiorly:** the gland lies on the inferior surface of the deep transverse perineal muscle
- 3 **Anteriorly:** the duct enters the ampulla urethrae
- 4 **Inferiorly:** the bulb of the penis is located under the glands



The prostate is the **largest male genital gland**. It produces **30 % of ejaculate fluid**. It is located **subperitoneally, inferior to the urinary bladder**. The **urethra courses through the prostate** along with the paired ejaculatory ducts, the common duct for the epididymis and the seminal glands.

External structure

- 1 **Base of prostate** (*basis prostatae*) – the broad cranial part of the prostate
 - is located around the neck of the urinary bladder
- 2 **Apex of prostate** (*apex prostatae*) – the tip of prostate, points towards the pelvic floor
- 3 **Prostatic urethra** (*pars prostatica urethrae*)
 - a segment of the urethra coursing through the prostate
- 4 **Anterior surface** (*facies anterior*) – faces the pubic symphysis, is attached to the pubic symphysis by the puboprostaticus muscle and puboprostatic ligaments
- 5 **Posterior surface** (*facies posterior*) – the posterior surface facing the rectum
- 6 **Right and left inferolateral surfaces** (*facies inferolateralis dextra et sinistra*)
 - the lateral surfaces facing the pelvic floor
- 7 **Right, left and middle lobes of the prostate** (*lobus dexter, sinister et medius*)
 - the three lobes of the prostate

Regional anatomy

- 8 **Cranially** – the neck of the urinary bladder, embedded in the base of the prostate
- 9 **Dorso-cranially** – the seminal glands laterally and the ductus deferens medially
- 10 **Centrally** – the urethra
- 11 **Dorsally** – the rectum, behind the rectovesical septum
- 12 **Caudally** – the apex of the prostate is embedded in the pelvic floor

Internal structure

- 1 **Prostatic capsule** – formed by three layers
 - 1.1 **Capsule of prostate** (*capsula prostatica*) – a fibromuscular double-layered capsule, tightly attached to the surface of the prostate
 - 1.2 **Prostatic venous plexus** (*plexus venosus prostaticus*)
 - 1.3 **Periprostatic capsule** (*capsula periprostatica*) – the visceral pelvic fascia
- 2 **Division of the prostate into histological zones** (according to McNeal)
 - 2.1 **Peripheral/inferoposterolateral zone** (*zona inferoposterolateralis/peripheralis*)
 - the outer part of the prostate, rich in glands
 - 2.2 **Central/superomedial zone** (*zona superomedialis/centralis*)
 - the internal wedge-shaped area around the openings of the ejaculatory ducts and the prostatic utricle
 - 2.3 **Peri-urethral gland zone** (*zona glandularum periurethralium*) – a slender strip of mucous glands in the wall of the urethra, the innermost zone
 - 2.4 **Transitional/anteromedial zone** (*zona anteromedialis/transitionis*)
 - is located between the peripheral and peri-urethral zones around the proximal part of the urethra
 - 2.5 **Anteriomedian zone / commisura** (*isthmus of prostate/zona anteromediana*)
 - a non-glandular zone composed of fibromuscular stroma
- 3 **Urethral crest** (*crista urethralis*) – an elevated crest on the posterior surface of the urethra
 - 3.1 **Seminal colliculus** (*colliculus seminalis*) – an elevated area on the urethral crest
 - 3.1.1 **Openings of the ejaculatory ducts** – on both sides of the seminal colliculus
 - 3.1.2 **Prostatic utricle** (*utricleus prostaticus*)
 - an unpaired blind remnant of the paramesonephric duct
- 4 **Prostatic sinus** (*sinus prostaticus*) – a pair of pits on both sides of the urethral crest where the prostatic ductules (ducts of prostatic glands) open

Vascular supply

Arterial supply: the prostatic branches of the inferior vesical artery and middle rectal artery

Venous drainage: the prostatic venous plexus and vesical venous plexus

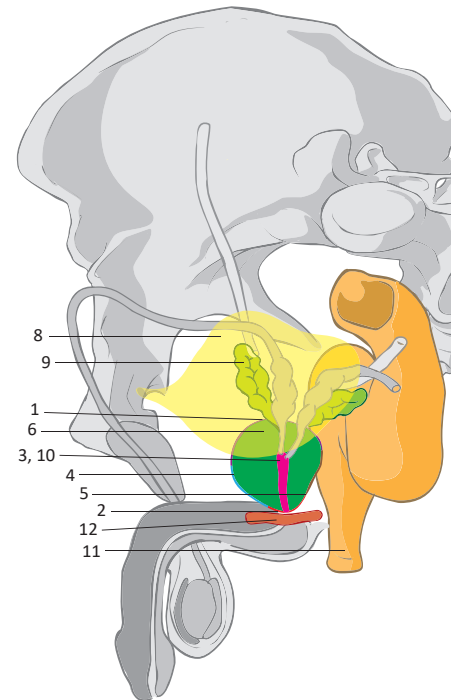
Lymphatic drainage: the internal and external iliac nodes and the sacral nodes

Innervation

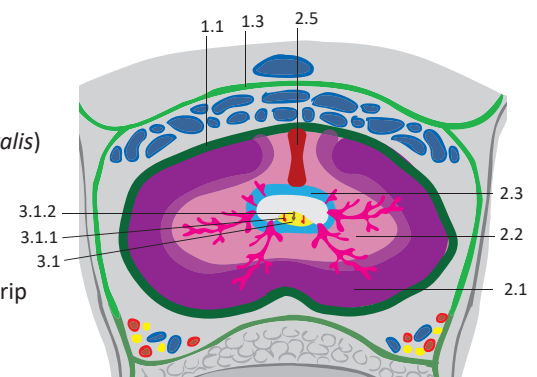
Sympathetic system: the prostatic plexus originating from the inferior hypogastric plexus

Parasympathetic plexus: the prostatic plexus originating from the inferior hypogastric plexus

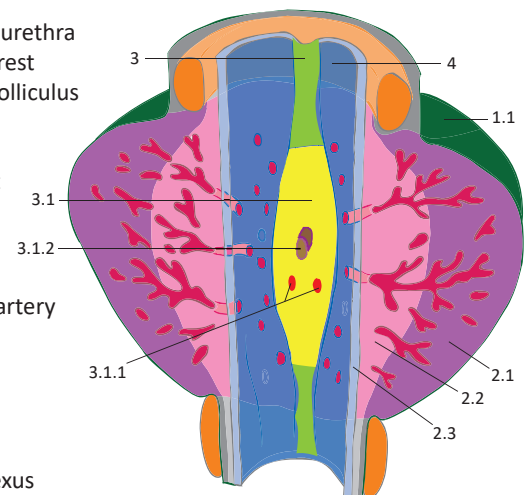
Viscerosensory system: nerve fibers in the parasympathetic and sympathetic prostatic plexus



Left superior-oblique view of the male pelvis



Transversal section of the prostate



Frontal section of the prostate, posterior view

The scrotum is a sac **located on the outside of the body**, which contains the testis and epididymis. The temperature inside the scrotum is **2–4 degrees lower than core body temperature**. This ensures optimal spermatogenesis. **Contraction of the cremaster muscle pulls the scrotum upwards, towards the perineum.**

External structure

Septum of scrotum (*septum scroti*) – a septum composed of connective tissue that separates the scrotum into two independent cavities, each containing one testis

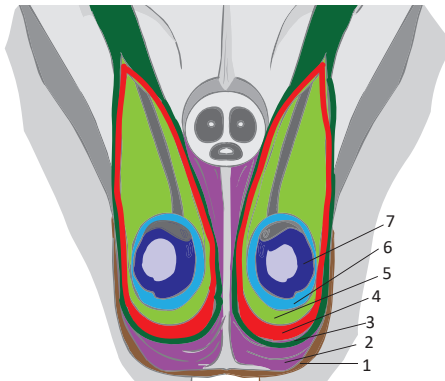
Raphe of scrotum (*raphe scroti*) – a seam-like line in the place where the genital (labioscrotal) swellings merged during development, right over the septum of the scrotum, – continues ventrally as the raphe of the penis and dorsally as the perineal raphe

Scrotal ligament (*ligamentum scrotale*) – a remnant of the gubernaculum, a cord of connective tissue attaching the lower pole of the testis and tail of the epididymis to the floor of the scrotum

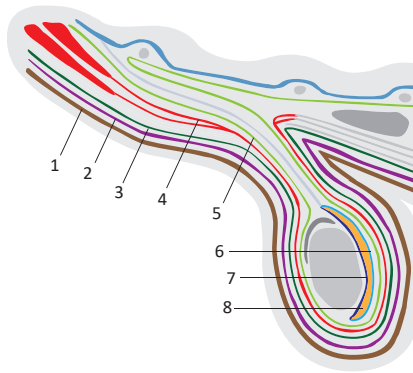
Layers of the scrotum

Layer of the scrotum	Corresponding layer of the abdominal wall
1 Skin	Skin
2 Dartos fascia (<i>tunica dartos</i>)	Fatty layer of superficial fascia
3 External spermatic fascia (<i>fascia spermatica externa</i>)	Membranous layer of superficial fascia
4 Cremaster and cremasteric fascia (<i>musculus cremaster, fascia cremasterica</i>)	Internal oblique Transversus abdominis
5 Internal spermatic fascia (<i>fascia spermatica interna</i>)	Transversalis fascia
6 Parietal layer of the tunica vaginalis (<i>periorchium</i>)	Parietal peritoneum
7 Visceral layer of the tunica vaginalis (<i>epiorchium</i>)	Visceral peritoneum

- 8 **Vaginal cavity of testis** (*cavitas vaginalis testis*)
– a serous cavity between the two layers of the tunica vaginalis



Anterior view of the layers of the scrotum



Schematic section through the layers of the scrotum and abdominal wall

Vascular supply

Arterial supply:

- **anterior half of the scrotum:** the anterior scrotal branches from the external pudendal arteries
- **posterior half of the scrotum:** the posterior scrotal branches from the internal pudendal artery
- **scrotal layers:** the cremasteric artery from the inferior epigastric artery

Venous drainage:

- the anterior scrotal veins, which drain into the great saphenous vein
- the posterior scrotal veins, which drain into the internal pudendal vein

Lymphatic drainage: the superficial inguinal nodes

Innervation

Motor system: the genital branch of the genitofemoral nerve

Somatosensory system:

- **anterior 1/3 of the scrotum:** the ilioinguinal nerve, genital branch of the genitofemoral nerve
- **posterior 2/3 of the scrotum:** the pudendal nerve, posterior cutaneous nerve of the thigh

The cremasteric reflex is triggered by tactile stimulation of the internal surface of the thigh, which leads to contraction of the cremaster. The afferent limb is formed by the somatosensory fibres of the femoral branch of the genitofemoral nerve. The efferent limb is formed by the motor fibres of the genital branch of the genitofemoral nerve.

Cavum serosum scroti is an obsolete term for the vaginal cavity of the testis.

The helicine arteries (*arteriae helicinae*) are convoluted in a flaccid penis and straighten during erection.

The tunica albuginea of the corpora cavernosa is denser than the tunica albuginea of the corpus spongiosum.

The bulbospongiosus is attached to the corpus spongiosum in the area of the bulb of the penis. It aids in emptying of the urethra and its rhythmic contractions contribute to the process of ejaculation.

Balanos is the Greek term for the bulb of penis. **Posthe** is the Greek term for foreskin.

Clinical notes

A hydrocele is an enlargement of the scrotum caused by accumulation of fluid inside the vaginal cavity of the testis. **A congenital hydrocele** results from a persistent communication between the peritoneal cavity and the layers of the scrotum via a patent vaginal process of peritoneum. **An acquired hydrocele** can occur as result of inflammation of the testis or epididymis or from a tumour. However, in the majority of cases, its origin remains unknown.

A scrotal hernia arises due to a persistent communication between the peritoneal cavity and vaginal cavity of the testis (or the vaginal process of the peritoneum). An intestinal loop enter the scrotum.

Cabanas' lymph node (*nodus lymphoideus inguinalis superficialis superomedialis*) is the sentinel node of the skin of the penis.

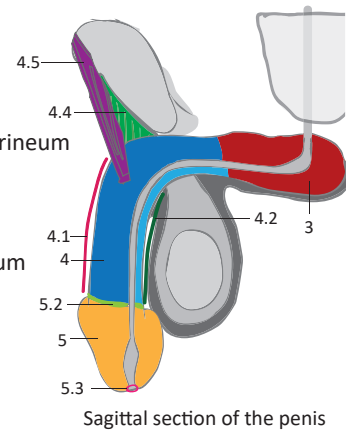
In a suspected testicular tumor, a diagnostic biopsy is contraindicated due to the risk of dissemination of tumour cells in the layers of the scrotum.

Balanoposthitis is inflammation of the mucosa of the bulb of the penis and the internal layer of the foreskin.

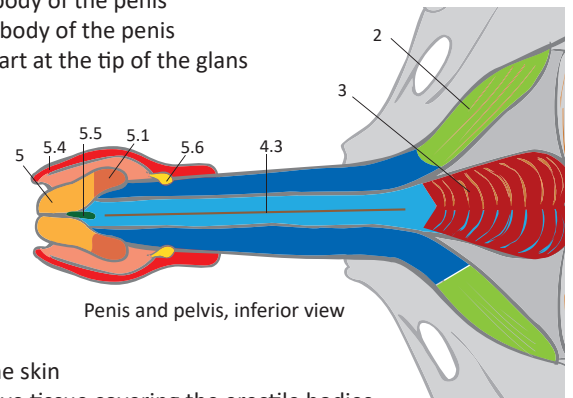
The **penis** is an external genital male organ with **both excretory and copulative functions**. It is formed from 3 erectile bodies and the urethra, which courses throughout its entire length.

External structure

- 1 **Root of penis** (*radix penis*) – composed of two crura
- 2 **Crura of penis** (*crura penis*) – a paired internal part of the corpora cavernosa
 - attaches to the phalic crest of the inferior pubic ramus and to the pubic symphysis
- 3 **Bulb of penis** (*bulbus penis*) – an unpaired dilated part of the corpus spongiosum underneath the perineum
- 4 **Body of penis** (*corpus penis*) – the external part of the penis, composed of three erectile bodies
 - 4.1 **Dorsum of penis** (*dorsum penis, facies dorsalis*) – the „anterior“ surface in a flaccid penis
in the anatomical position, the upper to posterior surface in an erect penis
 - 4.2 **Urethral surface** (*facies urethralis*) – the „posterior“ surface in a flaccid penis facing the scrotum
– the urethra courses the penis right beneath the urethral surface
 - 4.3 **Raphe of penis** (*raphe penis*) – a longitudinal seam-like line on the urethral surface
– a developmental remnant of the genital/labioscrotal swellings
 - 4.4 **Fundiform ligament of penis** (*ligamentum fundiforme penis*) – originates at the linea alba
– a loop-like ligament attaching the penis to the pubic symphysis
 - 4.5 **Suspensory ligament of penis** (*ligamentum suspensorium penis*)
– attaches the dorsum of the penis to the anterior surface of the pubic symphysis
- 5 **Glans penis** – the most distal segment of the corpus spongiosum, the location of the external urethral orifice
 - 5.1 **Corona of glans** (*corona glandis*) – a dilated rim of the glans facing the body of the penis
 - 5.2 **Neck of glans** (*collum glandis*) – a groove separating the glans from the body of the penis
 - 5.3 **External urethral orifice** (*ostium urethrae externum*) – the most distal part at the tip of the glans
 - 5.4 **Foreskin / prepuce** (*preputium*) – a skin fold of the body of the penis
that can be pulled over the glans to a variable extent
 - 5.5 **Frenulum** (*frenulum preputii*) – a ventrally located skin crease
– a transition between the skin of the glans and the skin of the prepuce
 - 5.6 **Preputial glands / Tyson's glands** (*glandulae preputiales*)
– small smegma-producing glands at the inner margin of the foreskin



Sagittal section of the penis



Penis and pelvis, inferior view

Internal structure

- 1 **Skin**
- 2 **Superficial fascia of penis** (*fascia penis superficialis*) – enables free sliding of the skin
- 3 **Deep fascia of penis / Buck' fascia** (*fascia penis profunda*) – a layer of connective tissue covering the erectile bodies
- 4 **Tunica albuginea** (*tunica albuginea*) – a dense layer of connective tissue covering the erectile bodies
- 5 **Corpora cavernosa penis** – erectile bodies that erect the penis when engorged with blood
 - the ischiocavernosus attaches to them, contributing to the process of ejaculation
 - 5.1 **Cavernous spaces and trabeculae of corpora cavernosa** – spaces that fill with venous blood during erection
– little cavities inside the erectile bodies lined with endothelium, fenestrated by trabeculae
 - 5.2 **Septum penis** – the incomplete fibrous plate separating the two corpora cavernosa
- 6 **Corpus spongiosum penis** – the unpaired erectile body beneath the urethral surface encompassing the spongy part of the male urethra; it constitutes the bulb and glans of the penis, and the body of the penis along with the corpora cavernosa

Vascular supply

Arterial supply: the internal pudendal artery

- 7 **Dorsal artery of penis** – supplies the skin and glans of the penis and the foreskin
- 8 **Deep artery of penis** – inside the corpus cavernosum, gives off the helicine arteries as its branches
- 9 **Artery of bulb of penis** – inside the bulb of the penis
- 10 **Urethral artery** – inside the corpus spongiosum

Venous drainage:

- 11 **Superficial dorsal vein of penis** – an unpaired vein that drains through the external pudendal veins into the common femoral vein
- 12 **Deep dorsal vein of penis** – an unpaired vein that bifurcates into the pudendal plexus of Santorini
(and further to the prostatic venous plexus), and into the internal pudendal vein which converges into the internal iliac vein

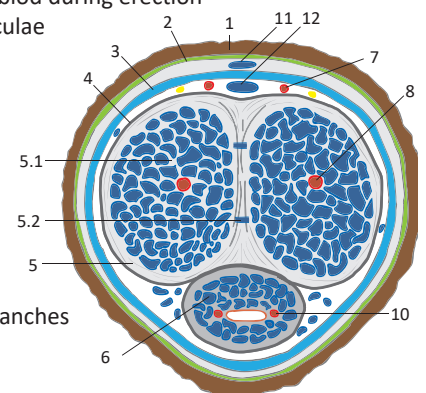
Lymphatic drainage: the superficial inguinal nodes; **glans of penis:** the deep inguinal nodes, external iliac nodes; **corpora cavernosa, corpus spongiosum, urethra:** the internal iliac nodes

Innervation

Somatosensory system: the ilioinguinal nerve (the skin of the root of the penis), dorsal nerve of the penis from the pudendal nerve

Sympathetic system: the cavernous nerves of the penis from the inferior hypogastric plexus

Parasympathetic system: the cavernous nerves of the penis from the sacral splanchnic nerves via the inferior hypogastric plexus



Transverse section of the penis

The male urethra is the **final segment of the male excretory and reproductive systems**. The sphincter vesicae prevents **retrograde flow of semen into the urinary bladder** and prevents the release of urine during ejaculation. Along its course, the male urethra contains **constrictions and dilations** and **two curvatures**.

External structure

Segments

- 1 **Internal urethral orifice** (*ostium urethrae internum*)
– the internal orifice inside the urinary bladder
- 2 **Intramural part** (*pars intramuralis, preprostatic part*) – a constricted part in the wall of the urinary bladder encircled by the smooth sphincter vesicae (*musculus sphincter vesicae*), which continues caudally as the smaller internal urethral sphincter
- 3 **Prostatic urethra** (*pars prostatica urethrae*) – a dilated part inside the prostate
– two ejaculatory ducts open at the seminal collicle, transporting semen into the urethra
- 4 **Membranous urethra / intermediate part of urethra** (*pars intermedia/membranacea*)
– a constricted part passing through the pelvic floor
– is surrounded by the striated external urethral sphincter
- 5 **Spongy urethra** (*pars spongiosa*) – passes within the corpus spongiosum of the penis
 - 5.1 **Ampulla urethrae** – the dilated initial segment, where the bulbo-urethral glands open
 - 5.2 **Navicular fossa** (*fossa navicularis*) – the dilated final segment of the urethra
- 6 **External urethral orifice** (*ostium urethrae externum*)

Curvatures

- 7 **Subpubic curvature** – a superior-facing concavity at the transition of the membranous and spongy urethra, impossible to straighten
- 8 **Prepubic curvature** – a inferior-facing concavity of the spongy urethra, straightens during erection

Sphincters

- 9 **Internal urethral sphincter**
– a weak smooth muscle sphincter innervated by visceromotor fibres of the sympathetic system
- 10 **External urethral sphincter** – a strong striated muscle innervated by somatomotor nerve fibres originating in the S2–S4 spinal cord segments

Constricted and dilated segments

Constricted segments

- 1 **Intramural part**
- 2 **Membranous urethra**
- 3 **Spongy urethra**
- 4 **External urethral orifice**

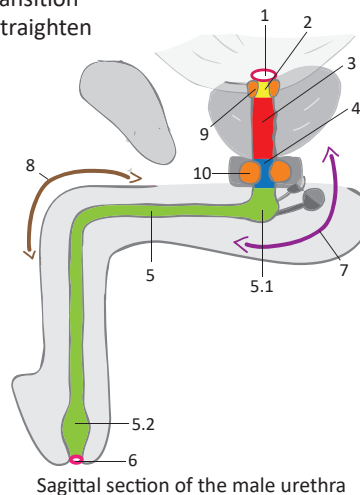
Dilated segments

- 5 **Prostatic urethra**
- 6 **Ampulla urethrae**
- 7 **Navicular fossa**

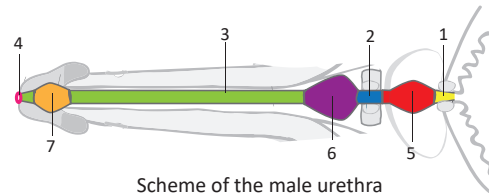
Mechanism of erection and ejaculation

Erection is triggered by the parasympathetic system and ejaculation by the sympathetic system.

1. **Sexual stimuli are processed by the hypothalamic nuclei** and the limbic system.
2. **The stimuli are transferred to spinal cord segment S3**, where the **parasympathetic sacral splanchnic nerves** originate and, as the cavernous nerves of the penis, innervate erectile bodies.
3. Under the influence of the parasympathetic system, **the intimal thickenings within the helicine arteries diminish**, which **increases blood flow to the cavernous spaces**.
4. **The erectile bodies enlarge and compress the cavernous vein of penis**, preventing blood outflow.
5. **The prepubic curvature straightens** and the **penis enlarges**.
6. Stimulation of sensory nerve endings in the penis leads to **activation of the ejaculation center in the spinal cord segments T11–L2** under the influence of the sympathetic system.
7. The sympathetic system **contracts the ductus deferens**, which transports spermatozoa from the epididymis to the prostatic urethra, where they mix with fluid from prostate and seminal glands.
8. Concurrently, under the influence of the sympathetic system, **the spincter vesicae and internal urethral sphincter contract**, **preventing retrograde flow of semen into the urinary bladder**.
9. **Contractions of the bulbospongiosus and ischiocavernosus eject semen out of the urethra.**



Sagittal section of the male urethra



Scheme of the male urethra

The **20 cm long male urethra** is about 5 times longer than the 4 cm long female urethra. However, it is narrower and curved.

The **epithelium of the male urethra** varies along its course:

The **intramural part and prostatic urethra** above the opening for the ejaculatory ducts are lined with transitional epithelium.

The **caudal part of the prostatic urethra** and membranous and spongy urethrae are lined with stratified columnar epithelium.

The **navicular fossa** is lined with non-keratinised stratified squamous epithelium.

The **external urethral orifice** is lined with keratinised stratified squamous epithelium.

According to the WHO 2010 data, normal values for physiological spermogram are as follows:

Semen volume > 1.5 ml

Spermatozoa concentration

> 15 million spermatozoa/ml

Spermatozoa motility > 40 %

Spermatozoa morphology > 4 %.

Clinical notes

Clinically, the urethra can be divided in two parts: the anterior and posterior parts. From the anatomical point of view, the posterior urethra consists of the intramural part, prostatic urethra and membranous urethra. The anterior urethra consists of the spongy urethra.

Compared to the female urethra, the male urethra is longer and narrower and therefore less prone to infections. However, due to its course, these infections tend to have a worse clinical manifestation.

After transurethral resection of the prostate (TURP) for benign hyperplasia of the prostate, the neck of the urinary bladder remains open during ejaculation. This results in retrograde flow of semen into the urinary bladder, causing the so called "dry ejaculation".

Epispadias is a rare atypical location of the external urethral orifice on the dorsum of the penis (1:30 000)

Hypospadias is a more common atypical location of the external urethral orifice on the ventral (urethral) surface of the penis (1:300).

Nitric oxide (NO) relaxes vascular smooth muscle. Drugs for erectile dysfunction, such as sildenafil-Viagra, increase the concentration of NO in the vascular beds of the erectile bodies.

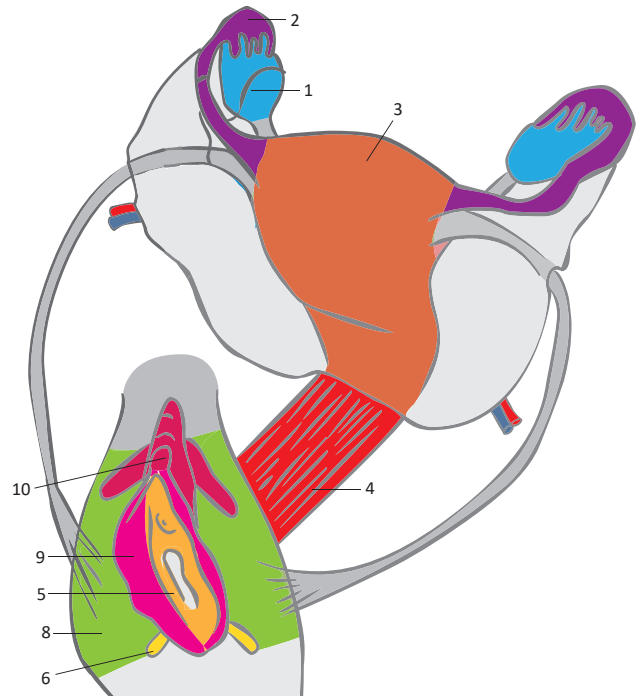
The internal female genital organs include the genital glands (ovaries), uterine tubes, uterus and vagina. The vestibule, labia, female erectile bodies (clitoris and bulb of the vestibule) and mons pubis constitute the external female genital organs.

Internal genital organs

- 1 **Ovary** (*ovarium*) – an intraperitoneally located pair of glands that contain maturing oocytes
- 2 **Uterine tube** (*tuba uterina*) – a paired tubular organ attached to the uterine horn
 - partially covers the ovary
 - the most common location of fertilisation of an oocyte
- 3 **Uterus** (*uterus*) – an unpaired hollow organ
 - houses the developing embryo and foetus
- 4 **Vagina** (*vagina*) – the terminal segment of the female genital system
 - the female organ of copulation and part of the birth canal

External genital organs

- 5 **Vestibule** (*vestibulum vaginae*)
 - an area containing the external openings of the vagina, urethra, and paired accessory glands
- 6 **Greater vestibular gland / Bartholin's gland** (*glandula vestibularis major Bartholini*)
 - a paired gland emptying into the vestibule
- 7 **Lesser vestibular glands** (*glandulae vestibulares minores*)
 - small glands emptying into the vestibule
- 8 **Labia majora** (*labia majora pudendi*)
 - cover the labia minora
- 9 **Labia minora** (*labia minora pudendi*)
 - cover the vestibule
- 10 **Clitoris** (*clitoris*) – the unpaired female erectile body
- 11 **Bulb of vestibule** (*bulbus vestibuli*)
 - a pair of female erectile bodies located around the vestibule



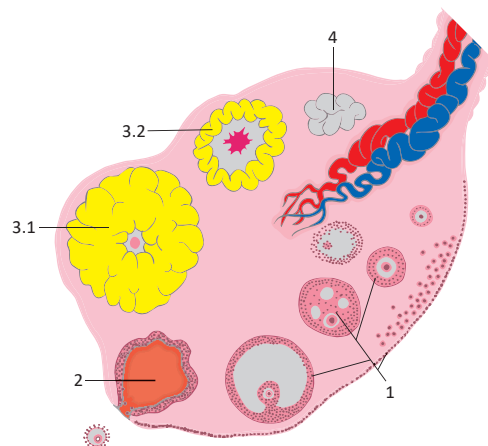
Left upper oblique view of the female genital system

Growth and maturation of follicle

Ovarian cycle – a 28 day cycle during which a single follicle reaches maturity
 – during the cycle, the ovarian cortex cells differentiate and form the layers of the ovarian follicle

Ovulation – a secondary oocyte is released from a mature follicle into the uterine tube where fertilisation takes place

- 1 **Maturation of the follicle**
 - several distinct stages of follicle maturation are distinguished:
 - 1.1 **Primordial follicle**
 - 1.2 **Primary follicle**
 - 1.3 **Secondary follicle**
 - 1.4 **Tertiary follicle**
 - 1.5 **Mature/Graafian follicle**
- 2 **Corpus rubrum/haemorrhagicum** (haemorrhagic body)
 - the immediate residue of a ruptured follicle after ovulation
- 3 **Corpus luteum**
 - develops from the corpus rubrum
 - secretes progesterone, estrogen, androgens, relaxin and inhibin
 - 3.1 **Corpus luteum of menstruation** (*corpus luteum menstruationis*)
 - is present during the second half of the menstrual cycle
 - 3.2 **Corpus luteum of pregnancy** (*corpus luteum graviditatis*)
 - produces progesterone until the 4th month of pregnancy
 - produces the polypeptide hormone relaxin, which loosens the soft tissue of the pubic symphysis and the sacro-iliac and sacrococcygeal joints, thus facilitating labour
- 4 **Corpus albicans**
 - a scar left after an obliterated corpus luteum of pregnancy

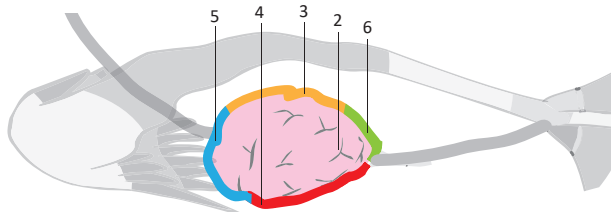


Stages of follicle maturation in the ovary

The ovary is the female genital gland (gonad). It is a paired intraperitoneal organ located in the **pelvis**. The ovary contains **oocytes within follicles, undergoing the process of maturation**. The ovary produces **various steroids and polypeptide hormones**.

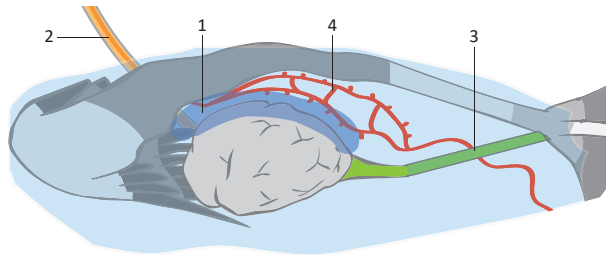
External structure

- 1 **Medial surface** (*facies medialis*) – faces the pelvic viscera
- 2 **Lateral surface** (*facies lateralis*) – lies on the lateral wall of the lesser pelvis
- 3 **Mesovarian border** (*margo mesovaricus*)
 - the anterior margin facing the peritoneal fold of the ovary
- 3.1 **Hilum of ovary** (*hilum ovarii*) – a slit through which the ovarian vessels and nerves enter the ovarian parenchyma
- 4 **Free border** (*margo liber*) – the posterior free margin
- 5 **Tubal extremity** (*extremitas tubaria*)
 - the upper pole facing the abdominal opening of the uterine tube
- 6 **Uterine extremity** (*extremitas uterina*) – the lower pole facing the uterine horn



Peritoneal folds and their corresponding arterial supply

- 1 **Mesovarium** – a peritoneal fold of the ovary
 - attaches to the posterior aspect of the broad ligament of the uterus
- 2 **Suspensory ligament of ovary** (*ligamentum suspensorium ovarii*)
 - attaches the tubal extremity of the ovary to the lateral pelvic wall
 - contains the ovarian artery and veins, ovarian nervous plexus and lymphatic vessels
- 3 **Ligament of ovary** (*ligamentum ovarii proprium*)
 - attaches the uterine extremity of the ovary to the uterine horns
 - contains the ovarian branch of the uterine artery and its corresponding veins and lymphatic vessels
- 4 **Ovarian arcade** – an anastomosis of the ovarian branch of the uterine artery with the ovarian artery



Internal structure

Surface – simple cuboidal epithelium

Tunica albuginea (*tunica albuginea ovarii*) – a thick layer of connective tissue under the epithelium

Ovarian stroma (*stroma ovarii*) – the connective tissue of the ovarian parenchyma

– forms the ovarian cortex and medulla

Ovarian cortex (*cortex ovarii*) – the functional tissue of the ovary

– contains the follicles in their various stages of maturation

Ovarian medulla (*medulla ovarii*) – contains the ovarian nerves and vessels and connective tissue

Vascular supply

Arterial supply: ovarian artery and ovarian branch of the uterine artery

Venous drainage: left ovarian vein (to left renal vein) and right ovarian vein (to inferior vena cava)

Lymphatic drainage: lumbar nodes

Innervation

Sympathetic system: ovarian, renal and aortic plexuses

Parasympathetic system: ovarian plexus (from the vagus nerve with accessory fibres from the superior and inferior hypogastric plexuses)

Viscerosensory system: sensory fibres travel in the sympathetic and parasympathetic plexuses

Oophoron is the Greek term for ovary.

Salpinx is the Greek term for uterine tube.

Ligamentum uteroovaricum and **chorda uteroovarica** are obsolete terms for the ligament of the ovary.

Ligamentum infundibulopelvicum is an obsolete term for the suspensory ligament of the ovary.

The adnexa of the uterus is a collective term for the ovary, uterine tube and their peritoneal folds.

Clinical notes

Oophorectomy is the surgical removal of the ovary. **Salpingectomy** is the surgical removal of the uterine tube.

Salpingo-oophorectomy / adnexectomy is the surgical removal of both the ovary and the uterine tube.

Adnexitis / pelvic inflammatory disease is inflammation of both the uterine tube and ovary. It may be caused by spread of infection from the appendix.

Salpingitis is inflammation of the uterine tube. It can spread to the peritoneum via the abdominal ostium and thus cause peritonitis.

An ovarian cyst is a collection of fluid covered by a thin membrane located in the ovary. In majority of cases, they are of no clinical significance. However, sometimes they can lead to infertility. In diagnostic ultrasound, small cysts less than 40mm may cause diagnostic confusion due to their resemblance of normal follicles.

Tubal obstruction may result from recurrent inflammation of the uterine tube mucosa and consequent scarring, causing adhesions and subsequent loss of tubal patency. This is a common cause of female infertility and ectopic pregnancy.

Ectopic pregnancy (*graviditas extrauterina / GEU*) occurs when a zygote (a fertilised ovum) implants in a different location than its normal implantation site within the uterus. This occurs most commonly within the uterine tube. However, other locations, such as the cervical canal, ovary or peritoneal cavity are possible. In the majority of cases the embryo does not survive. However sometimes its development may continue and rupture the uterine tube causing life threatening bleeding and sepsis.

The uterine tube (Fallopian tube) is a paired muscular tube uniting the uterine horn with the ovary. Its abdominal ostium is **freely open to the peritoneal cavity**, where it **captures released oocytes**, which are transported along the tube by peristaltic contractions to the uterine cavity. **Fertilisations of an oocyte by a spermatozoon** usually happens within the uterine tube.

External structure

- **1 Abdominal ostium** (*ostium abdominale*) – faces the ovary
 - directly communicates with the peritoneal cavity
 - the inlet of the uterine tube
- **2 Infundibulum** (*infundibulum tubae uterinae*) – a dilated part adjacent to the abdominal ostium
 - contains the fimbriae
 - **2.1 Fimbriae** (*fimbriae tubae uterinae*) – mucosal appendages projecting from the abdominal ostium
 - **2.2 Ovarian fimbria** (*fimbria ovarica*) – the most distinct and longest fimbria, reaches the ovary
- **3 Ampulla** (*ampulla tubae uterinae*) – the wider lateral 2/3 of the uterine tube, decorated by many mucosal folds
- **4 Isthmus** (*isthmus tubae uterinae*) – the narrower medial 1/3 of the uterine tube, contains scant mucosal folds
- **5 Uterine part** (*pars uterina*) – the intramural segment of the uterine tube, located inside the wall of the uterus
- **6 Uterine ostium** (*ostium uterinum*) – is located within the uterine horn
- **7 Mesosalpinx** (*mesosalpinx*) – the peritoneal fold of the uterine tube
 - **7.1** the cranial part of the broad ligament of the uterus
 - **7.2** contains an arterial anastomosis between the ovarian artery and the ovarian branch of the uterine artery (called the ovarian arcade) and its branches

Internal structure

Mucosa (*tunica mucosa*)

- simple columnar ciliated epithelium
- is organised in multiple folds, which decrease in number and thickness from the ampulla to the isthmus
- the epithelium contains ciliated cells, peg cells and basal cells
- the mucosal folds form the “mucosal labyrinth”, which is formed by primary, secondary and tertiary folds

Muscular layer (*tunica muscularis*)

- smooth muscle tissue organised into an external longitudinal layer and an internal circular layer

Serosa (*tunica serosa*)

- the external surface covered by peritoneum

Vascular supply

Arterial supply:

- **the lateral 1/3 of the uterine tube:**
the tubal branch of the ovarian artery (from the abdominal aorta)
- **the medial 2/3 of the uterine tube:**
the tubal branch of the uterine artery (from the internal iliac artery)

Venous drainage:

- the left ovarian artery (to the left renal vein)
- the right ovarian artery (to the inferior vena cava)
- the uterine venous plexus (to the uterine vein, internal iliac vein)

Lymphatic drainage:

- the major part of the uterine tube: the lumbar nodes
- the isthmus: the superficial inguinal nodes

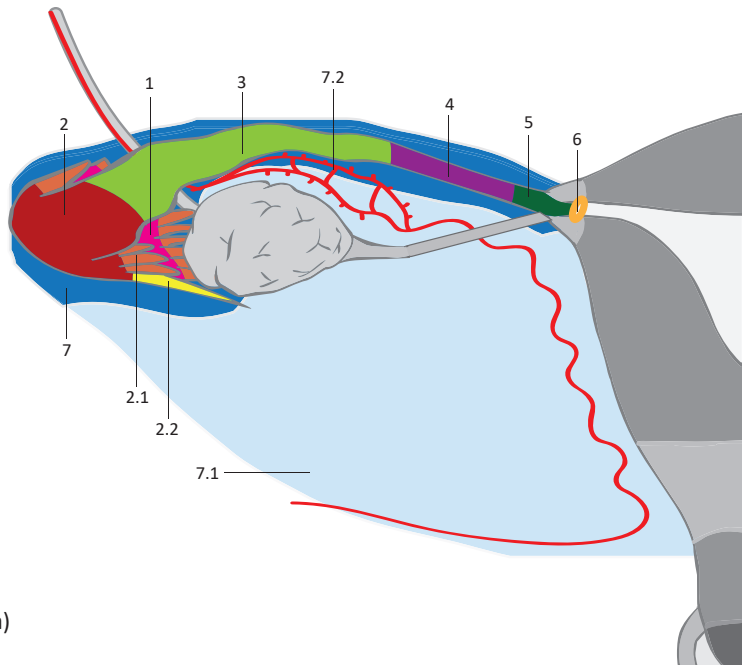
Innervation

Sympathetic system: the ovarian plexus (from the renal and aortic plexus)

Parasympathetic system:

- the lateral 1/3 of the uterine tube from the ovarian plexus (from vagus nerve)
- the medial 2/3 of the uterine tube from the inferior hypogastric plexus (from the sacral splanchnic nerves)

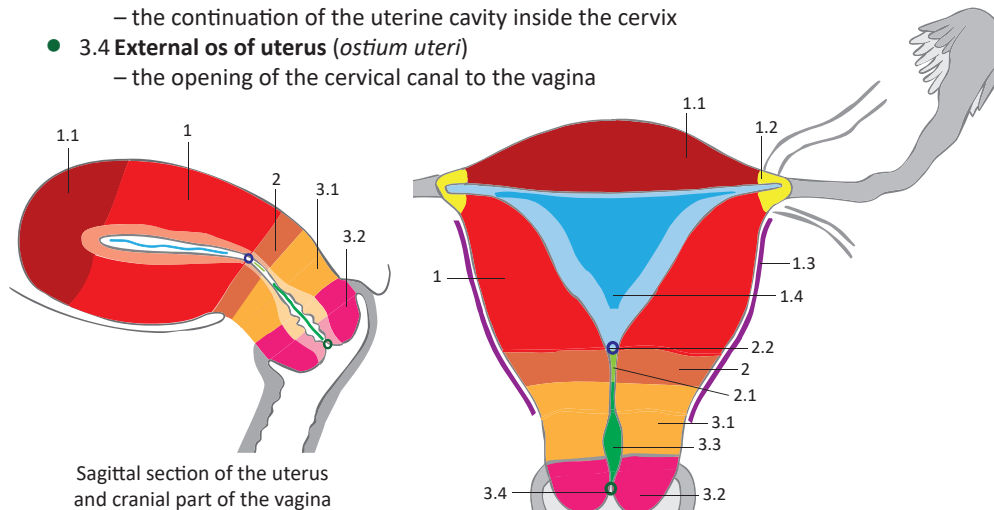
Viscerosensory system: sensory nerve fibres in the sympathetic and parasympathetic plexuses



The uterus is a **pear-shaped organ** located in the lesser pelvis between the urinary bladder and rectum. Its mucosa undergoes cyclic transformation, synchronised with the ovarian cycle. This ensures that the uterus is ready for implantation and the subsequent development of the fertilised ovum. It is positioned in **anteversion and anteflexion** and is almost completely covered by peritoneum. Due to its position, spermatozoa can easily reach the uterine cavity and it provides sufficient space for a growing embryo/foetus.

External structure

- **1 Body of uterus (*corpus uteri*)** – the middle and largest part of the uterus
 - **1.1 Fundus of uterus (*fundus uteri*)** – the ventro-cranial part of the body of the uterus
 - **1.2 Uterine horn (*cornu uteri*)** – a paired structure through which the uterine tubes enter the uterus
 - **1.3 Border of uterus (*margo uteri*)** – the attachment of the broad ligament of the uterus – the uterine artery ascends to the uterine horns from the isthmus of uterus
 - **1.4 Uterine cavity (*cavitas uteri*)** – houses the developing embryo and foetus
- **2 Isthmus of uterus (*isthmus uteri*)** – the narrow transitional area between the body and cervix
 - **2.1 Canalis isthmi** – the cavity of the isthmus, its upper border forms the internal os
 - **2.2 Anatomical os (*ostium anatomicum uteri internum*)** – the internal os
- **3 Cervix of uterus (*cervix uteri*)**
 - **3.1 Supravaginal part (*portio supravaginalis cervicis*)** – the cranial part of the cervix above the attachment of the vagina – contains mucosal folds called the palmate folds (*plicae palmatae*)
 - **3.2 Vaginal part (*portio vaginalis cervicis*)** – the caudal part of the cervix protruding into the vagina
 - **3.3 Cervical canal (*canalis cervicis*)** – the continuation of the uterine cavity inside the cervix
 - **3.4 External os of uterus (*ostium uteri*)** – the opening of the cervical canal to the vagina

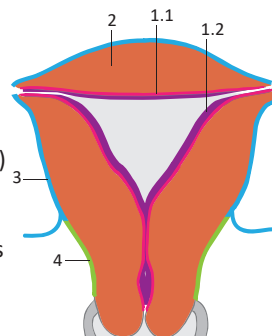


Sagittal section of the uterus and cranial part of the vagina

Frontal section of the uterus with the left uterine tube

Internal structure

- **1 Endometrium** – the uterine mucosa, consists of simple columnar epithelium, connective tissue and simple tubular glands, contains more goblet cells than ciliated cells
 - **1.1 Basal layer (*stratum basale*)** – the deepest layer of the endometrium
 - doesn't change during the menstrual cycle,
 - contains the basal parts of the endometrial glands
 - **1.2 Functional layer (*stratum functionale*)** – the widest layer
 - undergoes cyclic changes
 - is composed mainly of uterine glands
 - **1.3 Compact/superficial layer (*stratum compactum/superficiale*)**
 - the most superficial layer
 - contains the duct of the uterine glands
- **2 Myometrium** – smooth muscle of the uterus arranged in four layers
- **3 Perimetrium** – a serous coat of the uterus, continuous with the peritoneum of the broad ligament of the uterus
- **4 Parametrium**
 - loose connective tissue surrounding the caudal part of the uterus
 - extends between two layers of the broad ligament of the uterus and
 - below the broad ligament condenses to form the parametrial ligaments



Frontal section of the uterus

Metra and hystera are Greek terms for uterus.

The round ligament of the uterus extends from the uterine horn laterally towards the pelvic wall. It passes ventrally through the inguinal canal and inserts into the fibrous tissue of the labia majora. The ligament contains a few fascicles of smooth muscle and some vessels.

Menstruation is the phase of uterine bleeding that occurs at the end of the menstrual cycle. After menstruation, the uterine mucosa regenerates during the proliferative phase while an oocyte matures in the ovary. After the proliferative phase, the oocyte is released from a Graafian follicle into the uterine tube. This is followed by the secretory phase, which prepares the mucosa for implantation of a fertilized ovum. If fertilisation doesn't occur, menstruation occurs and the cycle repeats.

Clinical notes

Metrorrhagia is uterine bleeding that differs in timing or intensity to normal menstrual bleeding.

Metralgia is the Latin term for "pain of the uterus".

Hysterectomy is the surgical removal of the uterus.

The epithelium of the cervix changes during puberty. The epithelium becomes more vulnerable to infection with HPV (human papilloma virus) during this transformation. Infection with HPV can cause cervical carcinoma.

The vesicouterine fold (*plica vesicouterina*) is located at the bottom of the vesicouterine pouch. It is an important structure for surgeries such as hysterectomies and caesarean sections. An incision has to be made in the fold to release and descend the urinary bladder preventing possible surgical injury.

A myoma / leiomyoma is a benign tumor of the myometrium. It is the most common tumour of the body of the uterus. Its symptoms include long heavy bleeding that occurs independently from menstrual bleeding.

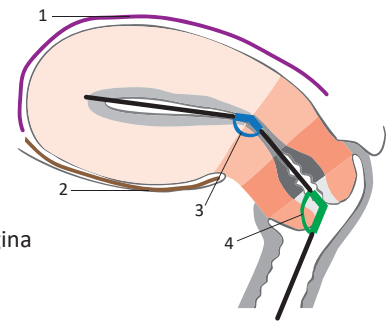
Uterine surfaces and position

Surfaces

- **1 Intestinal surface** (*facies intestinalis*) – the dorso-cranial surface facing the intestinal loops
- **2 Vesical surface** (*facies vesicalis*) – the ventro-caudal surface facing the urinary bladder

Position

- **3 Anteflexion** – the longitudinal axis of the body of the uterus and the longitudinal cervical axis form an obtuse angle of 160°–170° open ventrally with the vertex at the isthmus of the uterus
- **4 Anteversion** – the longitudinal axis of the body of the uterus and the longitudinal axis of the vagina form an angle of 70°–100° open ventrally with the vertex at the external os of the uterus
- **5 Dextroversion** – the uterus is often slightly turned to the right with its left border lying closer to the midline and pushed forward

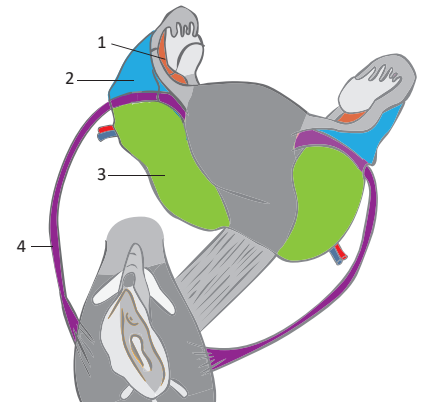


Sagittal section of the uterus and cranial part of vagina

Supporting apparatus of the uterus

The supporting apparatus of the uterus consists of the muscles of the pelvic floor. These are more important than the parametral ligaments.

- 1 Levator ani/pubovaginalis
- 2 Ischiococcygeus
- 3 Perineal muscles



Female genitalia, oblique view

Peritoneal folds

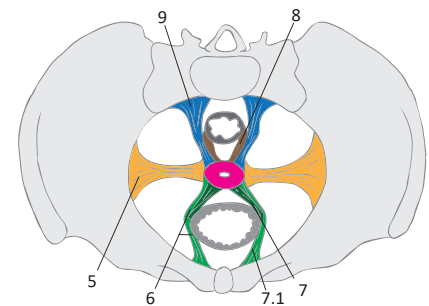
Broad ligament of uterus (*ligamentum latum uteri*) – is a double peritoneal fold located in the frontal plane of the lesser pelvis, consists of the three parts – it is attached to the uterine border medially and is laterally continuous with the parietal peritoneum of the pelvic wall

- **1 Mesovarium** (dorsally) – a peritoneal fold of the ovary
- **2 Mesosalpinx** (cranially) – a peritoneal fold of the uterine tube
- **3 Mesometrium** (caudally) – a peritoneal fold of the uterus

Parametrial ligaments

The parametrial ligaments hold the uterus in its proper position in the pelvis.

- **4 Round ligament of uterus** (*ligamentum teres uteri*) – keeps the uterus in anteversion during pregnancy
- **5 Cardinal ligament / Mackenrodt's ligament / transverse cervical ligament** (*ligamentum cardinale Mackenrodti / ligamentum transversum cervicis*) – a paired fibrous band extending laterally from the cervix and isthmus to the pelvic wall – enables antero-posterior movements of the uterus
- **6 Pubocervical ligament** (*ligamentum pubocervicale*) – extends from the pubic symphysis to the cervix
- **7 Vesico-uterine ligament** (*ligamentum vesicouterinum*) – extends from the uterus to the urinary bladder
 - **7.1 Pubovesical ligament** (*ligamentum pubovesicale*) – the continuation of the vesicouterine ligament to the pubis
- **8 Recto-uterine ligament** (*ligamentum rectouterinum*) – extends from the rectum to the uterus
- **9 Uterosacral ligaments** (*ligamenta sacrouterina*) – the continuation of the recto-uterine ligament to the sacrum, contain the inferior hypogastric plexus



Parametrial ligaments inside the pelvis

Blood supply

Arterial supply: the internal iliac artery

- **1 Uterine artery** – approaches the isthmus from the side and ascends tortuously along the uterine border

Venous drainage: the uterine venous plexus (into the uterine vein and internal iliac vein)

Lymphatic drainage:

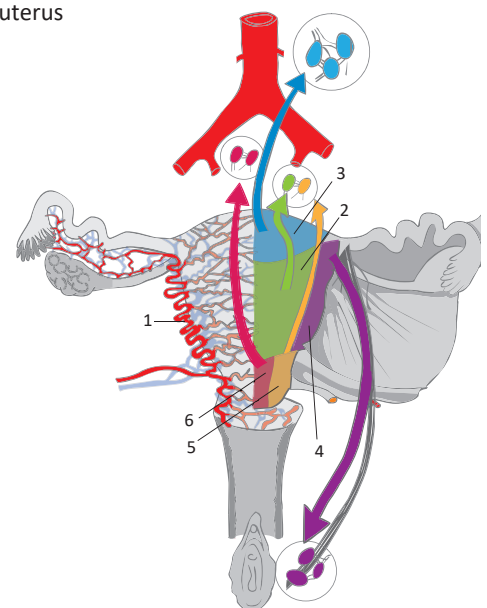
- **2 Body of uterus:** the internal iliac nodes
- **3 Fundus of uterus:** the lumbar nodes
- **4 Border of uterus:** the superficial inguinal nodes along the round ligament of the uterus
- **5 Ventral aspect of cervix:** the internal iliac nodes
- **6 Dorsal aspect of cervix:** the sacral nodes

Innervation

Sympathetic system: the uterovaginal plexus from the inferior hypogastric plexus

Parasympathetic system: the uterovaginal plexus from the inferior hypogastric plexus

Viscerosensory system: nerve fibers in both plexuses

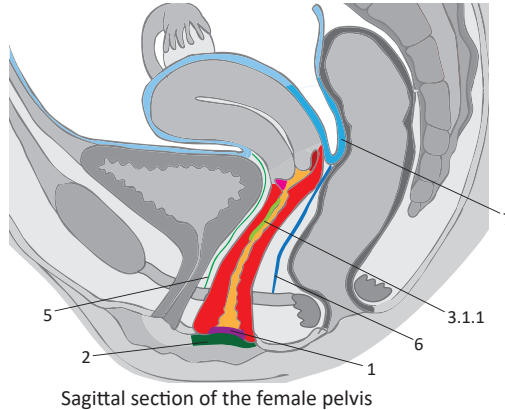
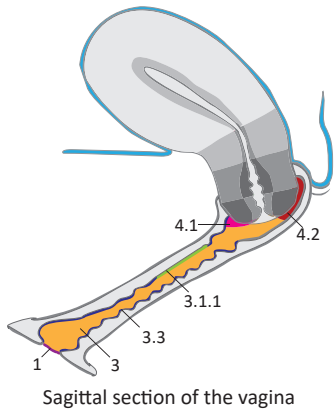


Blood supply of the female genitalia

The vagina is a **hollow muscular tube** that serves as the **terminal part of the female genital tract**. It facilitates **copulation and partuition**.

External structure

- 1 **Vaginal orifice** (*ostium vaginae*) – the external opening of the vagina in the vestibule
- 2 **Vestibule** (*vestibulum vaginae*) – a space bordered by the labia minora
- 3 **Vaginal canal** (*canalis vaginalis*) – the cavity of the vagina
 - 3.1 **Anterior wall** (*paries anterior*)
 - 3.1.1 **Pawlik's triangle** (*area trigonalis vaginae*) – an area on the anterior wall absent in rugae under the overlying trigone of the bladder
 - 3.2 **Paries posterior** (*paries posterior*)
 - 3.3 **Vaginal rugae** (*rugae vaginales*) – transverse mucosal folds on the vaginal walls – disappear during labour
 - 3.3.1 **Anterior and posterior vaginal columns** (*columna rugarum anterior et posterior*) – longitudinal elevations on the anterior and posterior walls of the vagina
- 4 **Vaginal fornix** (*fornix vaginae*) – encloses the vaginal part of the cervix uteri
 - 4.1 **Anterior part** (*pars anterior*) – the shallow anterior part of the fornix
 - 4.2 **Posterior part** (*pars posterior*) – the deeper posterior part of the fornix
 - 4.3 **Lateral parts** (*partes laterales*) – lateral parts of the fornix



Regional anatomy

- 5 **Urethrovaginal septum** (*septum urethrovaginale*) – a plate of connective tissue between the urethra and vagina
- 6 **Rectovaginal septum** (*septum rectovaginale*) – a plate of connective tissue between the vagina and rectum – the cranial extension of the perineal body (*corpus perineale*)
- 7 **Recto-uterine pouch / pouch of Douglas** (*excavatio rectouterina Douglassi*) – the most caudal recess of the peritoneal cavity in females, extends to the posterior fornix

Vascular supply

Arterial supply:

- the vaginal artery, which arises either directly from the internal iliac or from the uterine artery
- vaginal branches from the uterine artery, middle rectal artery and internal pudendal artery

Venous drainage: the internal iliac vein via the uterine venous plexus (into the uterine and vaginal veins), middle rectal and internal pudendal veins

Lymphatic drainage: internal and external iliac nodes, sacral nodes – caudally below the hymen: the superficial inguinal nodes

Innervation

Sympathetic system: the vaginal nerves, the uterovaginal plexus from the inferior hypogastric plexus

Parasympathetic system: the vaginal nerves, the uterovaginal plexus from the sacral splanchnic nerves (S2–S4)

Viscerosensory and somatosensory system: the vaginal nerves – caudally below the hymen: the pudendal nerve and its branches, inferior rectal nerves, perineal nerves, posterior labial nerves

Kolpos is the Greek term for vagina. **Introitus** is an obsolete term for the vaginal orifice.

The hymen is a mucous membrane separating the vestibule from the cavity of the vagina.

Mons pubis / mons Venus is a fat pad located above the pubic symphysis. It is covered with pubic hair, which grows under the influence of the female sex hormones.

The urethral carina of the vagina (*carina urethralis vaginae*) is a ridge on the anterior wall of the vagina formed by the underlying urethra.

The fascia of the clitoris (*fascia clitoridis*) covers the clitoris and attaches it to the anterior surface of the pubic symphysis.

The pubovaginalis (part of the levator ani) circles the vagina at the urogenital hiatus and creates the promontorium vaginae of Thomas.

Receptaculum seminis is an obsolete term for the posterior part of the vaginal fornix that serves to collect semen after coitus.

Clinical notes

Colpitis is inflammation of the vagina.

The recto-uterine pouch/pouch of Douglas is a clinically important place for palpation and evacuation of fluid that has accumulated inside the peritoneal cavity.

Colposcopy is an endoscopic examination of the cervix through the vagina.

Culdoscopy is an endoscopic examination of the recto-uterine pouch. The endoscope is inserted through the posterior fornix of the vagina.

Vaginal (exfoliative) cytology is a repeated cytological examination of the vaginal epithelium over the course of the menstrual cycle.

A recto-vaginal fistula is a pathological communication between the vagina and rectum that carries a very high risk of infection.

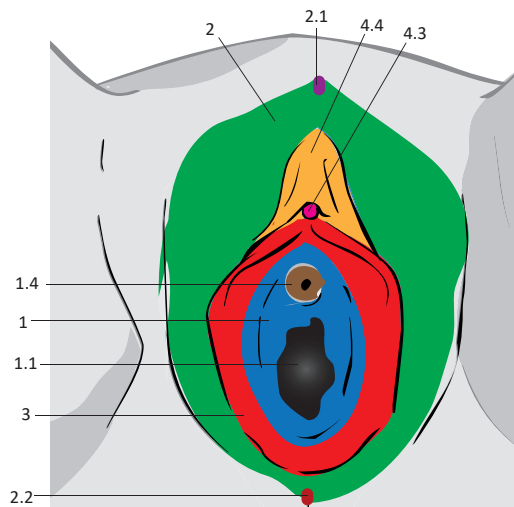
Local anesthetics can be applied in the area of the pudendal nerve during or shortly before labour.

The greater vestibular gland/Bartholin's gland can be a site of a painful abscess.

The **vulva** is a term for the female external genitalia. It includes the **labia majora**, **labia minora**, **clitoris**, **erectile bodies** and the **vestibule** (with its associated glands). The vestibule is an area bordered by the labia minora. The labia majora are covered with skin and pubic hair. The round ligament of the uterus inserts into the fat tissue of the labia majora. The labia minora do not contain fat tissue and are lined with mucosa on their medial surfaces. They divide ventrally into two folds, forming the prepuce and frenulum of the clitoris. The clitoris is an erectile organ connected to the two erectile bodies of the bulb of the vestibule.

External structure

- **1 Vestibule** (*vestibulum vaginae*)
 - **1.1 Vaginal orifice** (*ostium vaginae*) – the external opening of the vagina
 - **1.2 Greater vestibular gland / Bartholin's gland** (*glandula vestibularis major Bartholini*)
 - a paired mucous gland embedded in the perineal membrane
 - opens into the posterior part of the vestibule
 - **1.3 Lesser vestibular glands** (*glandulae vestibulares minores*)
 - **1.4 External urethral orifice** (*ostium urethrae externum*)
 - the external opening of the female urethra
- **2 Labia majora** (*labia majora pudendi*)
 - **2.1 Anterior commissure** (*commissura labiorum anterior*)
 - the anterior connection of the two labia majora
 - is located ventral to the prepuce of the clitoris
 - **2.2 Posterior commissure** (*commissura labiorum posterior*)
 - the posterior connection of the two labia majora, behind the vestibule
- **3 Labia minora** (*labia minora pudendi*)
- **4 Clitoris** – the female erectile organ
 - **4.1 Crura of clitoris** (*crura clitoridis*)
 - paired bodies of erectile tissue attached to the phallic crest of the inferior pubic ramus
 - meet in the midline at the body of the clitoris
 - **4.2 Body of clitoris** (*corpus clitoridis*) – is formed by fusion of the crura of the clitoris
 - **4.3 Glans of clitoris** (*glans clitoridis*) – the terminal part of the clitoris
 - **4.4 Prepuce of clitoris** (*preputium clitoridis*) – is located on the ventral aspect of the clitoris
 - **4.5 Frenulum of clitoris** (*frenulum clitoridis*)
- **5 Bulb of vestibule** (*bulbus vestibuli*) – a pair of erectile bodies encircling the vestibule
 - ventrally connected at the commissure of the bulbs (*commissura bulborum*)



Vascular supply

Arterial supply:

- the anterior labial branches (from external pudendal artery, which is from the femoral artery)
- the posterior labial branches (from internal pudendal artery, which is from the internal iliac artery)
- the deep artery of clitoris, dorsal artery of clitoris and artery of bulb of vestibule (from internal pudendal artery, which is from the internal iliac artery)

Venous drainage:

- the superficial dorsal vein of the clitoris and anterior labial veins (drain into the external pudendal veins, which drain into the common femoral vein)
- the deep dorsal vein of the clitoris (drains into the pudendal and vesical venous plexuses, which drain into the internal iliac vein)
- the posterior labial veins and vein of the bulb of the vestibule (drain into the internal pudendal vein, which drains into the internal iliac vein)

Lymphatic drainage:

- superficial inguinal nodes (bilaterally)
- deep inguinal nodes

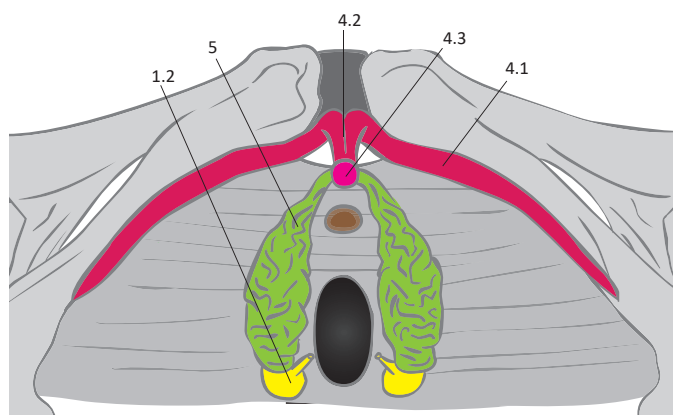
Innervation

Somatosensory innervation of the labia: the ilioinguinal nerve and the genitofemoral nerve (from the lumbar plexus)

Somatosensory innervation of the labia, clitoris and perineum: the pudendal nerve (from the sacral plexus)

Somatosensory innervation of the perineum: the posterior cutaneous nerve of thigh (from the sacral plexus)

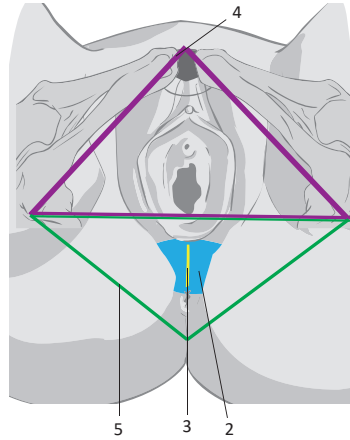
Visceromotor system of the clitoris: the cavernous nerves of the clitoris (from the inferior hypogastric plexus)



The perineum is the **region between the external genitalia and the anus**. The striated perineal muscles are arranged in two layers. They form the triangular fibromuscular **urogenital floor** (*diaphragma urogenitale*) spread between the ischial tuberosities and the inferior pubic rami. The urogenital floor contains a hiatus for the terminal segment of the urogenital tract. The perineal muscles **act as sphincters and contribute to erection and ejaculation**. They are **innervated by the pudendal nerve** (from spinal cord segments S2–S4).

Perineum

- 1 **In males:** the region between the posterior aspect of the scrotum and the anal canal
 - 2 **In females:** the region between the posterior commissure of the labia and the anal canal
 - 3 **Perineal raphe** (*raphe perinei*)
 - a skin ridge along the midline of the perineum
- Pelvic outlet** (*apertura pelvis inferior*)
- 4 **Urogenital triangle** (*trigonum urogenitale*)
 - the anterior triangle between the ischial tuberosities and pubic symphysis
 - contains the pelvic floor and perineal muscles
 - 5 **Anal triangle** (*trigonum anale*)
 - the posterior triangle between the ischial tuberosities and coccyx
 - the perineal muscles are missing here and the pelvic outlet is closed by the pelvic floor muscles



Urogenital floor in male

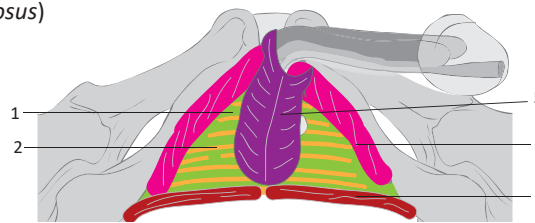
- 1 **Perineal membrane** (*membrana perinei*)
 - a fibrous sheet covering the inferior aspect of the deep transverse perineal muscle
- 2 **Deep transverse perineal muscle** (*musculus transversus perinei profundus*)
 - the deep muscular part of the urogenital floor
- 3 **Superficial transverse perineal muscle** (*musculus transversus perinei superficialis*)
 - forms the posterior border of the urogenital floor

Muscles located superficial to the perineal membrane

- 4 **Ischiocavernosus** (*musculus ischiocavernosus*)
 - compresses the crus of the penis, contributing to erection
- 5 **Bulbospongiosus** (*musculus bulbospongiosus*)
 - evacuates the urethra during micturition and ejaculation

Muscles located deep to the perineal membrane

- 6 **External urethral sphincter** (*musculus sphincter urethrae externus*)



Inferior view of the urogenital triangle in male

Urogenital floor in female

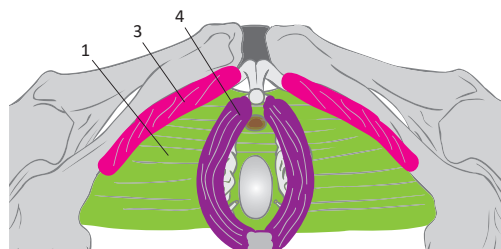
- 1 **Perineal membrane** – a fibrous sheet containing disperse muscle fibers
- 2 **Superficial transverse perineal muscle** (*musculus transversus perinei superficialis*)
 - adjacent to the posterior edge of urogenital floor, weak or often absent

Muscles located superficial to the perineal membrane

- 3 **Ischiocavernosus** – compresses the crus of the clitoris
- 4 **Bulbospongiosus** – compresses the vestibule, evacuating the vestibular glands

Muscles located deep to the perineal membrane

- 5 **External urethral sphincter**
- 6 **Compressor urethrae** (*musculus compressor urethrae*)
 - compresses and extends the urethra
- 7 **Sphincter urethrovaginalis** (*musculus sphincter urethrovaginalis*)
 - compresses and extends the urethra



Inferior view of the urogenital triangle in female

The **perineal body** (*corpus perineale*) (formerly known as *centrum tendineum perinei*) is a pyramidal fibromuscular structure located in the midline of the perineum between the urogenital and anal triangles. Cranially, the rectoprostatic septum / rectovaginal septum (*septum rectoprostaticum/rectovaginale*) continues from the perineal body. It is the central insertion of the perineal and pelvic floor muscles.

The **urogenital floor** (*diaphragma urogenitale*) is an obsolete term for the fibromuscular plate formed by the perineal membrane, transverse perinei muscles, ischiocavernosus and bulbospongiosus. In modern anatomy, this term is no longer used, however, it is frequently applied in clinical terminology, which is why we mention it in this text.

The **bulbo-clitoridal organ** is a clinical term for the continuous erectile tissue of cavernous bodies of clitoris, of bulb of vestibule including its commissure and of the intermediate part between these two.

The **pelvic floor** and its muscles page 131.

Lesser pelvis topography page 564.

Clinical notes

Episiotomy is a surgical incision of the perineum during labour to prevent uncontrolled injury of the vagina, perineum and the baby. Mid-line episiotomy crosses the perineal body, medio-lateral episiotomy (30° from the midline) cuts through the bulbospongiosus and the superficial transverse perineal muscle. Left medio-lateral episiotomy is easily sutured, heals faster and therefore is performed more often.

Cervical shortening, diagnosed by transvaginal ultrasound, is a sign of progressing labour. Obstetricians use it in women at risk of preterm labor. The cervix is of a rather conical shape in primigravida (a woman pregnant for the first time) and of a cylindrical shape in multigravida.

Pre-eclampsia is a clinical condition associated with maternal hypertension, proteinuria (presence of protein in urine) and oedema. This occurs most commonly due to histological changes in the uterine spiral arteries causing them to have higher blood pressure. Eventually, convulsions (eclampsia) may develop.

Pregnancy (*graviditas*) is the period of time from fertilisation to the extraction of the foetus/embryo from the uterus. Pregnancy terminates by either delivery or abortion. Giving birth to a child is defined as a **delivery** (*partus*). **Delivery of a live-born child** is defined if the neonate breathes and/or shows at least one vital sign. It is independent on the duration of pregnancy. **A stillborn child** is a foetus expelled or taken out of the uterus weighing over 500 grams and showing no signs of life. **A miscarriage** (*abortus*) is defined as the spontaneous expulsion of a foetus/embryo weighing less than 500 grams and showing no vital signs. An induced abortion is performed in accordance with local legislation.

Organ changes in pregnancy

1 Uterus

- the total weight of the uterus increases during pregnancy from 70 g to 1100 g (at term), due to increased blood flow (from 50 ml to 500–700 ml) and increased vasculature and fluid retention in the myometrium
- the total uterine volume increases from 10 ml to 5 litres (a 500 fold increase)

1.1 Cervix

- the dense fibro-muscular cervix acts as a firm plug for the uterus during pregnancy
- the cervix softens during labour; its collagen fibres break down and it retains fluid
- the cervix enables smooth passage of the foetus through the birth canal

1.2 Isthmus

- 1st trimester:** the length of the isthmus increases three-times and moves towards the body of the uterus
- the isthmus contributes to the formation of the lower uterine segment
- 2nd trimester:** the isthmus and body of the uterus become continuous
- the isthmus transforms into the lower uterine segment

1.3 Fundus – is palpable from the 12th week onwards

- 12th week** – palpable over the pubic symphysis
- 20th week** – palpable at the level of the umbilicus
- 36th week** – palpable under the xiphoid process

2 Vagina

- increased blood flow and fluid retention lead to enlargement and elongation of the vagina

3 External genitalia

- blood flow of the external genitalia increases
- pigmentation becomes more apparent
- submucosal veins dilate, sometimes forming varicosities
- the labia minora and the area around the external urethral orifice change color from pink to blue-like (livid); this is called Labhardt's sign

4 Round ligament of uterus and broad ligament of uterus

- both enlarge
- the broad ligament of the uterus enlarges due to an increase in number of the ovarian and uterine vessels and serves as a blood reservoir

5 Abdominal wall

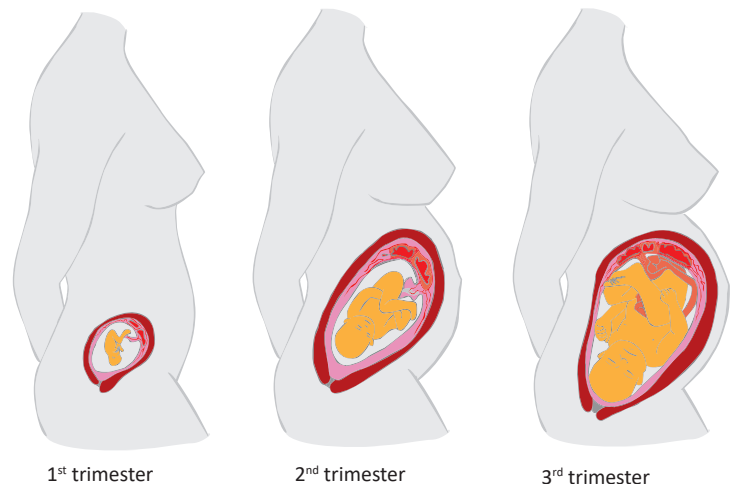
- pigmented spots appear on the skin (*striae gravidarum*)
- particularly in multiparous women, the rectus abdominis muscles may separate (diastasis), forming a slit through which the fundus may slip

6 Abdominal aorta and inferior vena cava

- in the supine position, the great vessels may be compressed by the growing uterus in the later stages of pregnancy
- compression of the more compliant inferior vena cava may result in the inferior vena cava syndrome
- inferior vena cava syndrome presents with hypotension (due to decreased venous return), nausea, fainting and foetal bradycardia on CTG (cardiotocogram)
- pregnant women are therefore positioned on their left side

7 Vertebral column – the cervical lordosis and the thoracic kyphosis equalise and the lumbar lordosis enhances

8 Joints – the hormone relaxin causes joints to loosen, particularly the sacro-iliac and sacrococcygeal joints and the pubic symphysis

1st trimester2nd trimester3rd trimester

Regional anatomy in pregnancy

- due to uterine enlargement, the abdominal organs are shifted to areas of minimal resistance

Cranially: jejunum, transverse colon

Cranio-laterally: ovary, uterine tubes

Vertical course: round ligament of uterus

Dorsally and to the right: the caecum, appendix

Laterally: ureter

The foetal membranes protect the developing human from external factors and form a stable environment inside the uterine cavity. The placenta provides a connection between the mother and the developing embryo/foetus. There are two main components of the placenta: the transformed uterine mucosa (*pars materna placentae*) and the extra-embryonic tissues and membranes of the developing embryo/foetus (*pars fetalis placentae*). The umbilical cord is an unpaired mobile connection between the foetus and the placenta.

Fetal membranes (*membrae fetales*)

1 Chorion

- originates from extra-embryonic mesodermal cells, is covered by trophoblasts
- differentiates into two functionally and morphologically distinct components:

1.1 Chorion frondosum / villous chorion

- is located at the insertion site of the placenta
- adjacent to the decidua basalis of the uterine mucosa
- is composed of chorionic villi that anchor the foetal part of the placenta into the uterine mucosa, the so called anchoring villi (*villi ancorantes*)
- the chorionic villi are rinsed with maternal blood providing exchange of oxygen, nutrients and waste products between the foetal and maternal circulation

1.2 Chorion laeve / smooth chorion

- a smooth larger part of the chorion lining the chorionic cavity underneath the amnion at the extra-placental sites
- adjacent to the parietal decidua (*decidua parietalis*) of the uterine mucosa of pregnancy

2 Amnion

- a single layer of cells differentiated from embryoblast
- the original amniotic cavity (*cavitas amniotica*) emerges as a slit in embryoblast, enlarges and eventually encloses the entire embryo (foetus), which swims inside this cavity filled with amniotic fluid

Placenta

1 Maternal part of the placenta (*pars materna placentae*)

- transformed stromal cells of the functional layer of the endometrium (decidual transformation); after fertilisation, under the influence of progesterone, they enlarge and fill with glycogen and lipids
- is separated into 15–20 lobules (cotyledons) by the placental septa growing from the basal (maternal) plate

1.1 Basal decidua (*decidua basalis*) – part of the decidual mucosa between the embryo and uterine wall

2 Foetal part of the placenta (*pars fetalis placentae*)

- forms the primary, secondary and tertiary villi

2.1 Amnion

2.2 Fetal mesenchyme with branches of umbilical vessels

2.3 Cytotrophoblast and syncytiotrophoblast

3 Placental layers (faces the uterine wall)

3.1 Amnion

3.2 Foetal mesenchyme and umbilical vessels

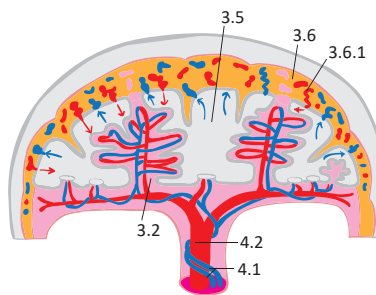
3.3 Cytotrophoblast

3.4 Syncytiotrophoblast

3.5 Intervillous space (*spatium intervillousum*)

3.6 Basalis decidua

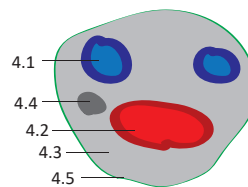
3.6.1 Spiral arteries



Cross-section of the placenta

Umbilical cord (*funiculus umbilicalis*) contents

- 4.1 Umbilical arteries (*arteriae umbilicales*) – two arteries
- 4.2 Umbilical vein (*vena umbilicalis impar*) – a single vein
- 4.3 Wharton's jelly (*textus mucoideus connectivus, mucoide connective tissue*)
- 4.4 Remnants of allantois and omphaloenteric ducts (*ductus omphaloentericus*)
- 4.5 Amnion – on the outer surface



Cross-section of the umbilical cord

According to the position of the embryo/foetus, we distinguish three parts of the uterine mucosa. The basal decidua is located right under the embryo on the future site of the placenta. The capsular decidua (*decidua capsularis*) surrounds the embryo. The rest of the uterine mucosa is the parietal decidua.

Vascular supply of the placenta:

Maternal blood from the spiral arteries fills the intervillous space and rinses the blood from the umbilical arteries in the chorionic villi. Functionally, the placenta is divided into small lobules (cotyledons) separated from each other by placental septa. A single terminal branch of the uterine artery supplies each cotyledon. The chorionic villi capillaries are in indirect contact through the wall of the villi with the maternal blood in the intervillous spaces. The umbilical vein drains foetal blood from the chorionic villi capillaries into the inferior vena cava of the foetus.

The gestational age of a child is calculated from the first day of the mother's last menstrual period (40 weeks at term). The fertilisation age is calculated from the day of fertilisation (38 weeks at term).

Embryology page 5.

Clinical notes

Caesarean section / C-section

(*sectio caesarea, S.C.*) is the surgical removal of a baby through the abdominal wall and uterus. It varies according to the surgical incision:

- lower uterine segment section
- cervico-corporal section in the body of the uterus and the lower uterine segment
- corporal section in the body of the uterus

Corporal section carries the highest risk of uterine rupture. Any following pregnancy must be terminated by C-section.

Placenta praevia a placenta located directly over the anatomical (internal) os of the uterus. Delivery must be performed by C-section as the placenta obstructs the birth canal.

Placenta accreta is a placenta growing to margin of the myometrium.

Placenta increta is a placenta invading the myometrium.

Placenta percreta is a placenta invading the entire layer of the myometrium or all three layers of the uterine wall.

Pregnancy (graviditas) is the **period of time from fertilisation to the end of labour**. Pregnancy lasts for 40 weeks. **Intrauterine** life of an individual is **divided into three periods** called trimesters. During labour, contractions of the uterine muscle and of the muscles of the abdominal wall lead to the expulsion of the fetus through the birth canal to the external world.

Clinically important anatomical markers of the foetus

- 1 **CRL**: crown-rump length
- 2 **GL**: global length
- 3 **NT**: nuchal translucency
- 4 **IT**: intracranial translucency
- 5 **NB**: (presence of) nasal bone

Important milestones of intra-uterine development

1st trimester (1st–14th week of pregnancy)

- all the organ systems are formed
- week 2**: implantation of the embryo, formation of the amniotic cavity, fluid and of the primary yolk sac
- week 3**: formation of three primary germ layers
- week 4 (day 20–22)**: first beats of the foetal heart
- week 4 (day 25–27)**: neural tube closure (anterior and posterior neuropore)
- end of week 8**: the end of the embryonal period, and the beginning of the foetal period

2nd trimester (14th–24th week of pregnancy)

- the foetus grows in length

3rd trimester (28th week till term)

- the foetus gains weight

Anatomy of labour

The mechanism of labour is a series of movements of the foetus as it progressing through the birth canal. These movements are caused by uterine contractions that force the foetus against the soft and hard tissues of the birth canal

1 **Progression of the head** (*progressio capitis*)

- after the cervix dilates, the head descends further into the pelvic floor, distends the perineum and passes out through the vulval ring
- after the head, the shoulders are delivered followed by the rest of the body
- 1.1 **Small fontanelle** (*fonticulus posterior*)
 - the leading point of head progression

● 2 **Initial head flexion** (*flexio capitis*)

- the head rotates so that its smallest diameter is in line with the transverse diameter (*diameter transversa*) of the pelvic inlet (*apertura pelvis superior*)

● 3 **Inner rotation** (*rotatio capitis interna*)

- 3.1 **to the oblique diameter** (*diameter obliqua*)
 - in the midpelvis (*amplitudo pelvis*)
- 3.2 **to the anteroposterior diameter** (*diameter recta*)
 - in the midpelvis (*angustia pelvis*)
- 3.3 **to the anteroposterior diameter** (*diameter recta*)
 - in the pelvic outlet (*apertura pelvis inferior*)

● 4 **Head extension** (*deflexio, extensio capitis*) as soon as the suboccipital part of the head reaches the symphysis, the head bends backwards (retroflexion)

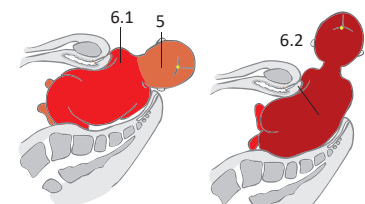
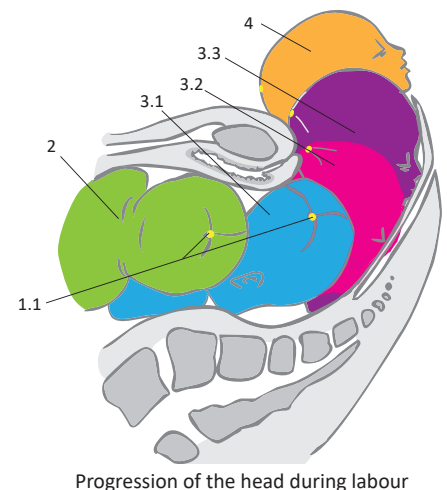
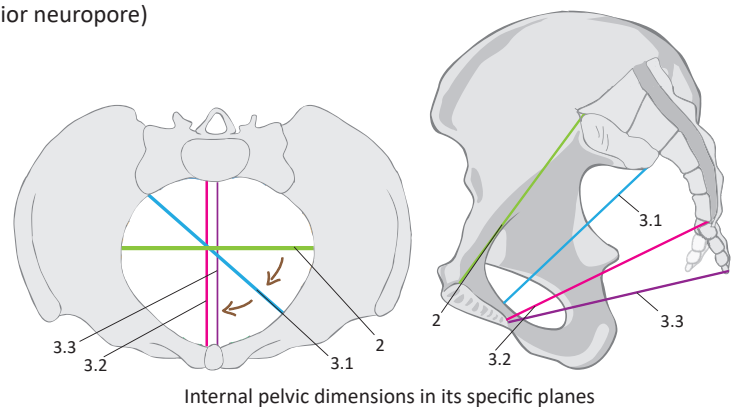
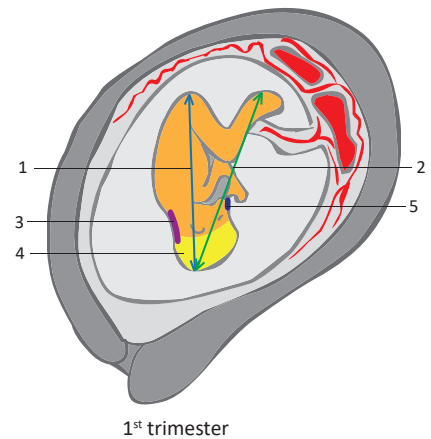
● 5 **External head rotation** (*rotatio capitis externa, restitutio*) the delivered head rotates so that it is in the same position as it was in the uterus

6 **Delivery of the shoulders** (*expulsio brachiorum*) follows after head delivery

- 6.1 **anterior shoulder** to half the length of the deltoid muscle
- 6.2 **posterior shoulder** fully delivered
- 6.3 the rest of the anterior shoulder is delivered

7 **Delivery of the trunk and lower limbs** does not follow a regular mechanism

8 **Delivery of the placenta** is commonly called the third stage of labour



I. Development of the genital system

1. State 3 structures originating from the mesonephric duct. (p. 238)
2. State 3 structures originating from the paramesonephric duct. (p. 238)
3. Explain which structures in the female are developmentally analogous to the root and raphe of the penis and to the spongy urethra. (p. 238)
4. Explain which structures in the male are developmentally analogous to the round ligament of the uterus and ligament of the ovary. (p. 238)

II. Male genital system

5. State 3 structures located in the scrotum. (p. 239)
6. State which serous cavity can only be found in male. (p. 239)
7. State the name of the gland that produces the major portion of ejaculate. (p. 239)

III. Testis

8. State the 2 layers of the tunica vaginalis and how they form. (p. 240)
9. State which 2 types of ducts can be found in the lobules of the testis. (p. 240)
10. Explain the main steps of the descent of the testis and why they occur. (p. 241)
11. Specify the latest age by which the testes should be fully descended and explain why. (p. 241)
12. Describe the lymphatic drainage of the testes and explain its clinical significance. (p. 241)
13. Explain the function and location of the Sertoli and Leydig cells. (p. 241)
14. Draw a scheme of the arterial supply and venous drainage of the testes. (p. 241)

IV. Epididymis

15. Describe the 3 parts of the epididymis and 3 ligaments that attach the epididymis to the scrotum. (p. 242)
16. State the structure that connects the duct of the epididymis with the prostate and urethra. (p. 242)
17. Specify the functional differences between the testis and epididymis in the process of spermatozoa maturation. (p. 242)

V. Ductus deferens and spermatic cord

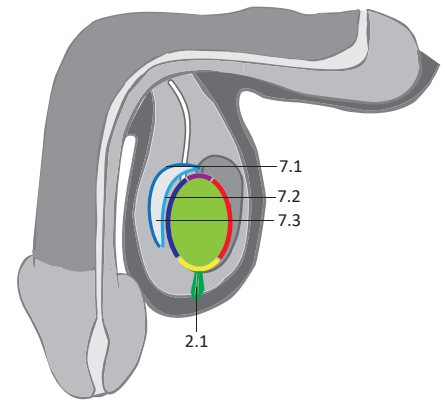
18. Explain the course of the six segments of the ductus deferens. (p. 243)
19. State which vessels pass posteriorly to the inguinal part of the ductus deferens. (p. 243)
20. List the 7 structures constituting the contents of the spermatic cord. (p. 243)
21. List the layers of the abdominal wall, eventually forming the layers of the spermatic cord. (p. 243)

VI. Seminal glands and prostate

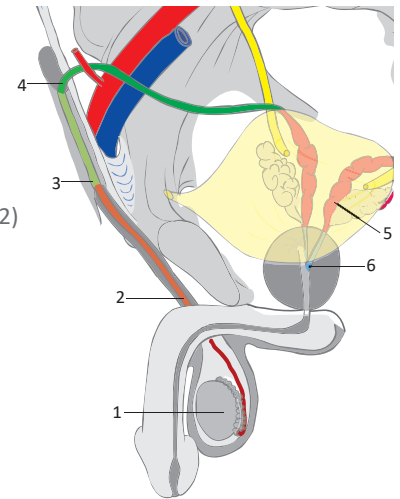
22. Describe the relation of the seminal glands to the ampulla of the ductus deferens. (p. 244)
23. State and describe the histological zones of the prostate and explain their clinical significance. (p. 245)
24. Explain the principle of the per rectum examination. (p. 245)
25. State the term used for the part of the prostate containing no glands. (p. 245)

VII. Scrotum

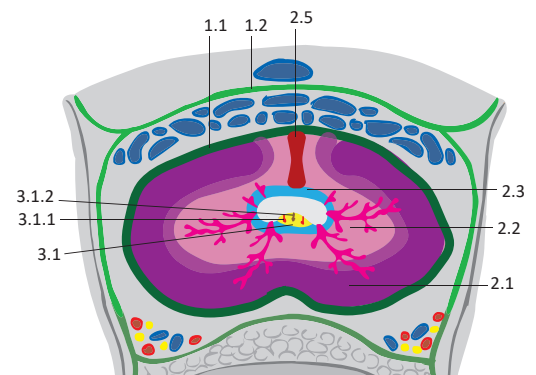
26. Draw a cross-section of the wall of the scrotum and describe its layers. (p. 246)
27. State the layers of the abdominal wall that form the cremaster muscle and fascia. (p. 246)
28. State the layer of the scrotum that originates from Scarpa's fascia. (p. 246)
29. Draw a scheme of the vascular supply and the innervation of the scrotum. (p. 246)
30. Explain the function of the genitofemoral nerve in the innervation of the scrotum. (p. 246)



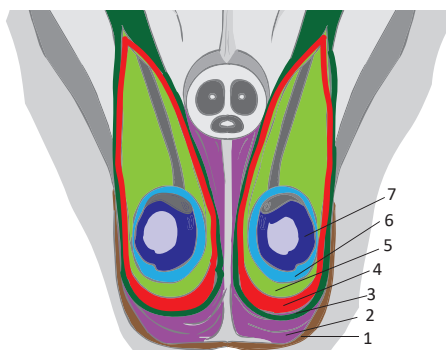
Describe the surfaces and parts of the testis



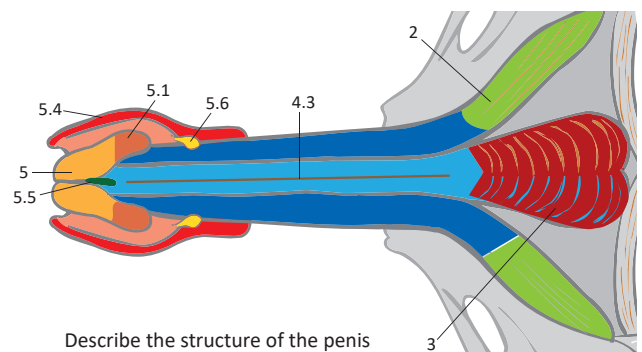
Describe the course of the ductus deferens



Describe the internal structure of the prostate



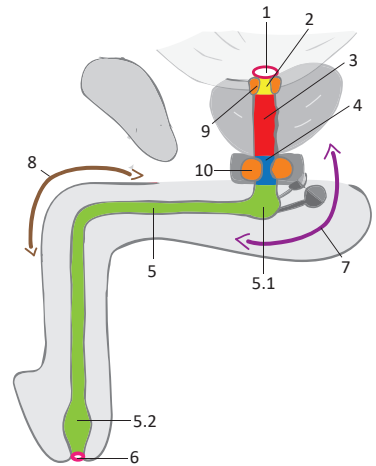
Describe the layers of the scrotum



Describe the structure of the penis

VIII. Penis and male urethra

31. In a longitudinal section, describe the five basic parts of the penis. (p. 247)
32. Draw a transverse section of the penis and describe its vascular supply and venous drainage. (p. 247)
33. List the constricted segments of the male urethra. (p. 248)
34. Explain whether and how it is possible to straighten the *curvatura prepubica*. (p. 248)
35. In a longitudinal section, describe the physiological dilations of the male urethra. (p. 248)
36. Explain the difference in the activity of the parasympathetic and sympathetic systems in process of erection and ejaculation. (p. 248)



Describe the parts of the male urethra

IX. Female genital system

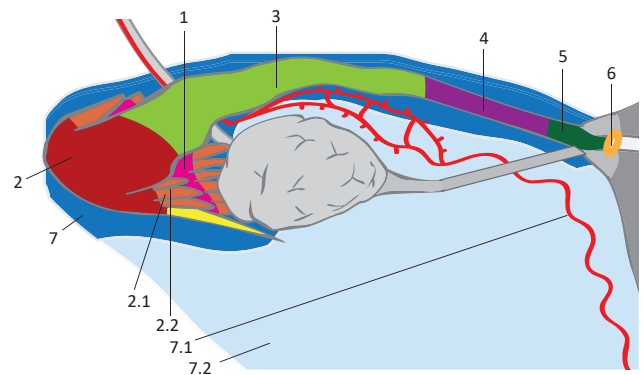
37. Name the internal and external female genital organs. (p. 249)
38. State the term for the final phase of development of a follicle before ovulation. (p. 249)
39. Explain the difference in function of the *corpus luteum menstruationis* and *corpus luteum graviditatis*. (p. 249)

X. Ovary

40. Describe the ligaments and the peritoneal fold of the ovary using a figure. (p. 250)
41. State the vessels that form the ovarian arcade. (p. 250)
42. State the layer of the ovary which contains its functional tissue. (p. 250)

XI. Uterine tube

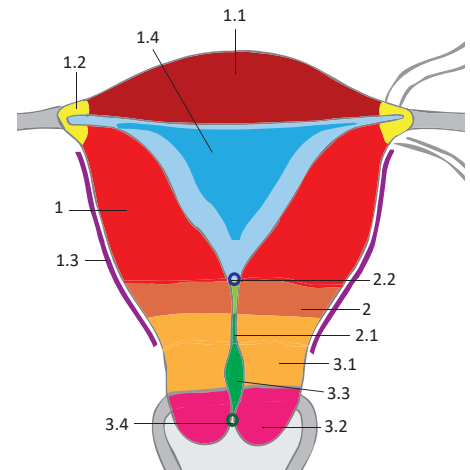
43. Name the six parts of the uterine tube. (p. 251)
44. Explain the vascular supply of the uterine tube using a scheme. (p. 251)
45. Explain what is an ectopic pregnancy and state its most frequent location. (p. 251)



Describe the external structure of the Fallopian tube and its parts

XII. Uterus and vagina

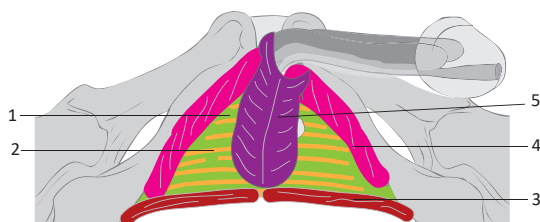
46. Describe the parts of the uterus using a figure. (p. 252)
47. Explain the difference between the *zona basalis* and *zona functionalis* of the endometrium. (p. 252)
48. Draw a sagittal section of the uterus in anteversion and anteflexion and specify their angles. (p. 253)
49. Describe the course of the *ligamenta cardinalia Mackenrodti*. (p. 253)
50. Draw a scheme of the lymphatic drainage of the uterus and specify the lymph nodes involved. (p. 253)
51. State the most caudal part of the peritoneal cavity in the female, how can it be palpated? (p. 254)
52. List the arteries supplying the vagina. (p. 254)



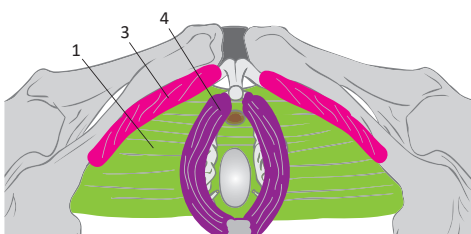
Describe the structure of the uterus

XIII. Female external genitalia and perineum

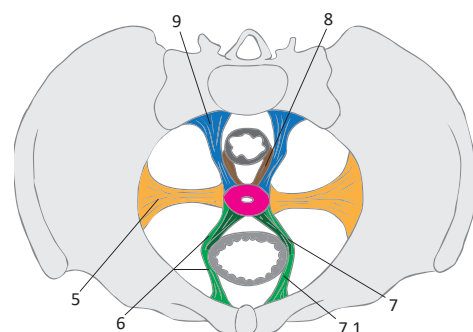
53. Specify the orifices found in the vestibule. (p. 255)
54. Name the 5 parts of the clitoris. (p. 255)
55. Describe the perineal muscles in male and female. (p. 256)



Describe the trigonum urogenitale in male



Describe the trigonum urogenitale in female



Describe the uterine parametral ligaments

We would like to **thank the following anatomists, clinicians and medical students** for their invaluable help, devotion and feedback in the preparation of this chapter.

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